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Regulatory cooperation and foreign portfolio investment☆

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ABSTRACT

We investigate the effect of cross-border regulatory cooperation in the enforcement of securities laws on global-mutual-fund portfolio allocations. Our research design exploits a shock to the Securities and Exchange Commission's oversight of foreign firms cross-listed on a US stock exchange around the signing of the Multilateral Memorandum of Understanding (MMoU), a non-binding, information-sharing arrangement between global securities regulators. In signatory countries, foreign investment in US-cross-listed firms increases by $110 billion relative to non-cross-listed firms. The strongest effects are for investors facing greater information asymmetries, those from countries closely linked to the US, and non-US foreign investors, suggesting significant spillover effects from international regulatory cooperation.

1. Introduction

In countries where domestic funding sources are limited, foreign portfolio investment is a critical source of financing (Bekaert et al., 2002). However, limited regulatory oversight and weak shareholder protection frequently discourage foreign investment in the countries most in need of external capital (Leuz et al., 2009). To overcome weak home-country institutions and increase their appeal to investors, firms often commit to stricter monitoring by cross-listing in a country with greater regulatory oversight, most commonly the US (e.g., Stulz, 1999). However, for many years, cross-border enforcement actions were rarely observed and the extent of the incremental oversight provided by a US cross-listing was uncertain (Siegel, 2005). More recently, there has been a notable expansion in extraterritorial enforcement by US regulators and an increase in cross-border cooperation between securities regulators (Silvers, 2016; 2018; Christensen et al., 2019).

In this study, we examine the impact of this increase in US enforcement and regulatory cooperation on investment in US cross-listed firms in the context of the International Organization of Securities Commission's (IOSCO) Multilateral Memorandum of Understanding (MMoU), the most comprehensive and rigorous standard of information sharing and cooperation to date. Following the events of September 11, 2001, IOSCO developed the MMoU as part of an effort to more effectively share information

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and facilitate enforcement cooperation between securities regulators across countries. The US Securities and Exchange Commission (SEC) was among the first to sign the MMoU in 2002 and prior research shows that cross-border SEC enforcement increases after countries sign the MMoU (Silvers, 2018).

We use the staggered adoption of the MMoU to investigate two questions. First, how does an increase in regulatory cooperation with the SEC affect foreign portfolio investment in US cross-listed firms? Second, which investors alter their holdings most following the increase in SEC oversight accompanying the MMoU? By examining the preferences revealed through changes in holdings, we aim to infer the investor-perceived benefits of regulatory oversight.

From a research design perspective, the MMoU provides a powerful setting to identify the effects of cooperative regulatory arrangements on foreign portfolio investment. From 2002 to 2017, securities regulators from over one hundred countries joined the arrangement. These staggered adoption dates allow us to better isolate the effect of the MMoU from other changes in investor preferences. Second, because SEC enforcement efforts under the MMoU focus on US-cross-listed firms, we can isolate the effect of the enforcement shock by comparing changes in ownership in US-cross-listed firms to domestic-only listed firms for the same funds in the same country. This approach controls for a wide range of factors that could affect investment levels, including macroeconomic shocks, changes in investor preferences, and other regulatory changes at the country level (e.g., Mulherin, 2007). Third, by using holdings measured at the fund-firm-quarter level, we can explore cross'sectional variation in the effect of the MMoU on mutual-fund investment based on characteristics of the MMoU signing country, fund country, type of investor, stock exchange, and target firm. If some investors perceive greater benefits from increased SEC oversight than others (e.g., foreign investors), their relative holdings of affected firms should increase following the signing of the MMoU.

Looking first at the effect of signing the MMoU on total investment (i.e., by both foreign and local funds), we show a statistically and economically significant 6.7 percentage point increase in the tilt towards US-cross-listed firms relative to non-US-cross-listed firms in the same country. Given an average total mutual fund investment in our sample of $1.6 trillion, this represents a shift of $110 billion in investment towards US-cross-listed firms (a 12% increase). Funds increase both their existing holdings (i.e., the intensive margin) as well as the number of cross-listed stocks in which they invest (i.e., the extensive margin). An analysis of counterfactual treatment effects in the pre-MMoU period suggests that the parallel-trends assumption is reasonable. Consistent with increased SEC oversight being a primary driver of the observed increase in investment, there is a significant increase in funds' ownership of firms cross-listed on a major US stock exchange [i.e., Level II and III American Depositary Receipts (ADRs)], though there is no evidence of an increase in investment in US-cross-listed firms traded over the counter (i.e., Level I ADRs).

Next, we compare the effect of the MMoU on local funds to its effect on foreign funds, where foreign and local refer to the location of the investor relative to the domicile of the investee. The literature on home bias indicates that geographically proximate investors have an information advantage over non-local investors. Furthermore, Baik et al. (2010) provide evidence of informational advantages for local investors within the US, which implies that the local information advantage comes from sources other than (or in addition to) publicly disclosed financial reports. The additional regulatory oversight facilitated by the MMoU has the potential to reduce the information advantage of local investors by extending the reach of US securities law with respect to issues such as insider trading, market manipulation, and financial reporting quality. As a result, foreign investors should benefit more from the MMoU than do local investors, even for investments made in relatively more transparent ADR firms.

Consistent with the prediction that foreign investors benefit most from the MMoU, we find that foreign mutual funds increase their investment in US-cross-listed stocks significantly more than do local mutual funds after a country signs the MMoU. Compared with their pre-MMoU preference for US-cross-listed shares, foreign funds increase their portfolio tilt toward ADRs by 6.9 percentage points. Consistent with the MMoU reducing the relative information disadvantage faced by foreign investors, foreign funds likely to suffer from the greatest information asymmetry with respect to the investee country (as captured by the two countries being geographically farther apart, having different languages, and having a high degree of distrust toward one another) appear to place more value on the additional oversight. Similarly, splitting countries based on their preexisting ties to the US, the effect of the MMoU on foreign investment is particularly pronounced in signatory (i.e., investee) countries without a preestablished working relationship with the SEC and with weak (pre-MMoU) economic ties to the US. Finally, splitting based on funds' investment strategies, dedicated (i.e., low turnover) funds, which prior research suggests are more likely to hold concentrated positions based on firm-specific information, respond more strongly to the MMoU than do transient (i.e., high turnover) funds.

To assess the potential spillover effects of the increase in SEC oversight that accompanies the MMoU, we next partition foreign investors into US and non-US funds. While the primary mandate of the SEC is to protect US investors, because SEC oversight is a public good, there could also be benefits of SEC oversight for non-US investors. Moreover, if US investors have less of an information disadvantage than other foreign investors prior to the MMoU, the benefits to foreign non-US investors could be even larger than for US investors. Consistent with this argument, prior research shows that because of their scale, resources, and substantial research capacity, US institutional investors tend to have informational advantages relative to other foreign investors (e.g., Albuquerque et al., 2009; Ferreira et al., 2017). If public monitoring substitutes for private information, non-US, foreign institutions might benefit at least as much as US investors from additional SEC oversight.

Consistent with a significant spillover benefit of SEC oversight to non-US investors, when partitioning foreign investors into US and non-US funds, our results indicate that non-US investment in US-cross-listed firms increases by 7.5 percentage points after the signing of the MMoU compared to an increase of 5.1 percentage points for US investors. Among non-US investors, there likely exist differences in the awareness of, and confidence in, the efficacy of the US government in general and the SEC in particular. We find that following adoption, the tilt toward US cross-listed firms by non-US foreign investors is stronger for funds from countries whose policy objectives are more closely linked to the US, as reflected by membership in the North Atlantic Treaty Organization (NATO) and similarities in United Nations (UN) voting behavior.

Finally, we provide descriptive evidence on how funds adjust their portfolios to reach their desired post-MMoU allocations. There are three notable possibilities, from highest to lowest transactions costs: (1) reallocating investments across countries (i.e., from non-MMoU countries to MMoU countries); (2) reallocating investments within a country (i.e., from non-ADR to ADR shares); and (3) disproportionately redirecting net fund flows to US-cross-listed shares in MMoU-signing countries. Shifting investment across countries is likely the most costly approach because it affects country-level target exposures in addition to transactions costs. Shifting within country avoids changes to country-level exposures but incurs transactions costs on both the purchase and sale. Redirecting fund flows is cheapest because there need not be any incremental transactions costs (i.e., purchases or sales are necessitated by fund flows). Funds likely start with the approach that has the lowest transactions costs and move progressively toward more costly approaches. The analysis of the portfolio reallocation costs that funds are willing to bear to achieve their desired post-MMoU portfolio allocations also provides an alternative way to assess the economic significance of our results.

Our results indicate that the observed increase in the tilt toward US-cross-listed firms is driven by investors reallocating funds within an MMoU-signing country, from non-ADR firms to ADR firms. After signing the MMoU, the level of foreign mutual fund investment in a signatory country stays relatively constant (i.e., funds do not significantly reallocate investments from non-MMoU to MMoU countries). Furthermore, when net fund flows in an investor country are positive (negative), funds allocate more (divest fewer) assets to US-cross-listed firms in MMoU-signing countries as compared to non-cross-listed firms. One interpretation of these results is that while the benefits of the additional regulatory oversight provided by the MMoU are large enough that funds significantly increase investments in US-cross-listed firms within signatory countries, they are not so large that funds change their country-level portfolio allocations.

The findings in this paper make several contributions to the literature. First, they provide the first (to our knowledge) evidence of cross-border spillovers from increased regulatory cooperation and information sharing. As noted in Leuz and Wysocki (2016), the literature lacks evidence on "market- or economy-wide effects from regulation, such as externalities, information spillovers and/or network effects." Our evidence suggests that networks of cooperative regulatory arrangements that increase US oversight abroad can have substantial spillover effects for non-US investors and firms.

Second, this paper provides novel evidence on the effects of increased US regulatory oversight on the geographic mix of a firm's investors. Prior work that evaluates the MMoU's impact on stock market outcomes like liquidity (Silvers, 2018) cannot separate the effect of greater enforcement across investor groups and consequently cannot speak to relative investor preferences or regulatory spillovers. Given the importance of foreign capital for economic growth and development, it is important to understand how changes in regulatory oversight affect investors' willingness to invest abroad. This paper shows that the MMoU increases foreign investors' preference for investing in firms that are subject to the increased regulatory oversight, particularly those investors facing a significant information disadvantage.

Third, this paper provides evidence on the effects of legal bonding by cross-listing (Stulz, 1999; Coffee, 1998; Licht, 1998; Benos and Weisbach, 2004). Prior research (e.g., Siegel, 2005) questions the extent to which the benefits associated with cross-listing result primarily from increased regulatory oversight. The MMoU provides a context where oversight changes while potential confounding factors, such as firm-cross-listing choice, trading location, changes in expected growth, and legal standing, are held relatively constant. Our results support the bonding hypothesis: stronger US oversight of cross-listed firms leads to an increase in foreign investment in those firms.

2. Institutional background and prior literature

Our primary goal is to assess how increased regulatory cooperation affects foreign portfolio investment, and, in particular, 1) how regulatory cooperation with the US SEC affects the willingness of foreign institutional investors to invest in US-cross-listed firms, and 2) what types of investors most prefer additional SEC oversight. We investigate these questions in the context of the staggered adoption of IOSCO's MMoU across 26 sample countries between 2002 and 2015. This section describes the MMoU's background, provisions, and implementation based on Silvers (2018), and discusses the relevant prior literature.

The MMoU framework was established in May 2002 and dictates information-sharing obligations for signatory countries, including the extent of the information shared, the permissible uses of this information, and any subsequent confidentiality obligations. As such, the MMoU is designed to cultivate cooperation across different jurisdictions and stipulates the standards to which regulators should adhere to facilitate cross-border cooperation. The MMoU has been more successful fostering information sharing than previous attempts because it was made a priority by many major economies and enjoyed support from a range of countries, which likely reflects the increased demand for information sharing following the terror attacks of the early 2000s. Prior to the MMoU, cross-country regulatory cooperation was limited to ad hoc investigations or support established through bilateral agreements. These approaches were fraught with difficulties and made little progress toward expanding cross-border enforcement capabilities. In contrast, Silvers (2018) finds that the MMoU significantly increased regulatory cooperation and cross-border enforcement actions.[1]

As shown in Table 1 Panel B, although major, developed countries began signing the MMoU in 2002, there is substantial variation in adoption timing. This reflects differences across countries in the process of deciding to join the MMoU, navigating the legislative protocol, establishing the required infrastructure, and obtaining IOSCO approval. Membership in the MMoU provides the right to vote within IOSCO and is a factor in financing decisions made by the International Monetary Fund (IMF) and the Financial Stability Board. The SEC participates actively in information sharing under the MMoU. The most common enforcement cases include insider trading, financial reporting improprieties, and bribery under the Foreign Corrupt Practices Act.[2]

There is limited research examining the effects of regulatory oversight in general, or the MMoU in particular, on the global mix of a firm's investors. Leuz et al. (2009) show that foreigners invest less in firms from countries that provide poor protection for outside investors. Kho et al. (2009) argue that the optimal level of insider ownership is higher in countries with weak (outside) investor protection, and thus, these countries have lower foreign ownership. More directly related to our setting, Ferreira and Matos (2008) examine the determinants of global institutional ownership and find that foreign holdings are higher for cross-listed firms.

However, since firms self-select into cross-listing based on variables such as investor demand and growth options and because there are many other factors associated with cross-listing (such as the type of institution permitted to invest in the stock), it is difficult to attribute the ownership structure of cross-listed firms to the effects of regulation. The MMoU represents a shock that is outside a firm's control and is therefore less susceptible to self-selection concerns. Additionally, the MMoU setting allows for better identification of the effects of regulatory oversight on investor clienteles because it provides a within-country control group that is less likely to be affected by the change in oversight and because adoption dates are staggered across countries over time (e.g., Mulherin, 2007). Despite its potential importance, to our knowledge, no prior paper examines the effects of regulatory cooperation on foreign portfolio investment.

3. Research design

In our primary analyses, we focus on the effect of the MMoU on foreign portfolio investment in US-cross-listed firms as compared to non-cross-listed firms. This focus is helpful for several reasons. First, cooperative regulatory arrangements are designed (in part) to facilitate cross-border investment. Cross-listed firms provide an obvious setting where the MMoU could enhance extraterritorial enforcement by the SEC.

Second, because foreign investors typically face an information disadvantage and higher monitoring costs than do local (i.e., geographically proximate) investors, strong regulatory oversight is likely to be particularly important for foreign investors (Leuz et al., 2009). Cross-listed firms typically have greater financial reporting transparency (e.g., English-language financial statements and more analyst coverage), and thus foreign investors could face less information asymmetry in these firms. However, cross-listing is unlikely to completely mitigate local investors' information advantage. For instance, local investors can observe a firm's public disclosures in addition to their own private information, some of which is likely derived from sources that are not directly related to public disclosure (e.g., local language media reports, conversations with firm insiders, political connections, and privileged access to the local legal system). Through such sources, local investors can gain access to better information about misreporting, insider trading, related-party transactions, self-dealing, fraud, etc. Consistent with this idea, Eleswarapu and Venkataraman (2006) find that country-level institutional features affect the liquidity of cross-listed firms in ways that are consistent with the bonding mechanism being incomplete.

If the MMoU increases regulatory oversight, and thus decreases the likelihood that a firm engages in any of the aforementioned activities, it can reduce local investors' information advantage. Consistent with this argument, Baik et al. (2010) find that, even within the US, local investors have an information advantage over non-local investors, which implies that local investors' information advantage comes from sources besides firm-specific (publicly disclosed) financial reporting information. Similarly, Jagannathan et al. (2017) find that fund managers with local expertise attract more investment and earn higher returns than do other fund managers.

3.1. Identification strategy

The primary empirical challenge in this paper is separating investment changes driven by a preference for the increased SEC oversight accompanying the MMoU from changes caused by variation in economic conditions or other non-regulation-based factors that affect foreign investors' preferences (Mulherin, 2007). In this regard, the MMoU setting has several advantages. Specifically, because the effect of the MMoU is likely to be strongest for US-cross-listed firms, firms from the same country without a US cross-listing provide a natural control group for changes in country-level macroeconomic conditions and foreign investors' general preferences for a given country at a particular time (e.g., contemporaneous country-level changes in domestic regulations). A disadvantage is that, if there are spillover effects of increased US regulatory oversight on non-cross-listed firms, this research design biases against detecting the effect of the MMoU.

[1] In addition, between 2003 and 2017, there were more than 27,000 information requests between regulators under the MMoU (https://www.iosco.org/about/?subsection=mmou).

[2] In 2018 alone, the SEC reported that it made 1,210 requests for information from foreign regulators and received 684 requests from foreign regulators. Because the SEC does not report separate statistics for requests made under the MMoU or provide detail on specific information requests, it is impossible to know what fraction of these requests were made pursuant to the MMoU (https://www.sec.gov/files/secfy20congbudgjust_0.pdf).

While a within-country comparison mitigates many identification concerns, it is important to consider other potential confounds not explicitly addressed by this design (e.g., that the timing of the regulations' implementation could be correlated with other, non-MMoU-based changes in investor preferences for cross-listed firms). Several other features of the MMoU setting help mitigate these concerns. First, countries sign on to the MMoU in a staggered fashion over more than a decade and the exact adoption timing is largely driven by country-level differences in the protocol for navigating the legislative process, establishing the required infrastructure, and obtaining IOSCO approval. Thus, it is unlikely that the adoption timing is correlated with short-term changes in economic circumstances that could affect investment preferences (e.g., Christensen et al., 2016).

Second, the focus on foreign investment allows us to test a variety of cross-sectional predictions about the investor countries, investee countries, and firms most likely to be affected by enhanced cross-border oversight. To the extent that the results from these cross-sectional comparisons are consistent with our predictions, they provide further support for our inferences and give insight into the specific mechanisms underlying the observed effects of the MMoU.

Third, controlling for changes in profitability, capital structure, and liquidity ensures that these factors, or unobservable factors correlated with these firm characteristics, do not drive the empirical results. Finally, the analysis of pre-MMoU trends in foreign investment for cross-listed and non-cross-listed firms provides no indication of significant differences in investment trends across these two groups, suggesting that the parallel-trends assumption in our difference-in-differences analysis is reasonable (see Section 4.1 for further details).

3.2. Regression model

Our primary regression analysis examines changes in mutual fund ownership in US-cross-listed firms following the MMoU signing in the investee firms' home country relative to changes for the same funds investing in non-US-cross-listed firms in the same country (i.e., we measure changes in investor preferences for cross-listed stocks in a given country relative to their overall preferences for stocks from that country). Specifically, our measure of foreign investment is the aggregate (for each quarter and for each fund-country/investee-firm-country pair) ratio of investment in US-cross-listed firms to total investment. Formally, for investee-firm-country i, fund'country j, and quarter t, U_t is the set of all firms cross-listed on a US exchange and V_t is the set of all firms that are not cross-listed on a US exchange. We then calculate:

$$\frac{XLInv_{ijt}}{Inv_{ijt}} = \frac{\sum_{m\in j}\sum_{(n\in i)\cap(n\in U_t)} Inv_{mnt}}{\sum_{m\in j}\sum_{(n\in i)\cap[n\in(U_t\cup V_t)]} Inv_{mnt}}, \tag{1}$$

where m denotes a mutual fund, n denotes an investee firm, and Inv_{mnt} is investment by fund m in firm n's equity securities (Stocks or ADRs) during quarter t.[3] In our regressions, we multiply this fraction by 100 so that the results can be interpreted in percentage points. To measure the effect of the MMoU signing, we estimate the following regression model using Ordinary Least Squares (OLS):

$$\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100 = \alpha_1 PostMMoU_{it} + \beta Controls_{it} + \gamma_{ij} + \delta_{jt} + \varepsilon_{ijt}. \tag{2}$$

By aggregating the dependent variable to the fund-country/firm-country-pair level in each quarter, we effectively create a measure of the country-specific investment for a value-weighted fund of funds from a given investor country. To avoid disproportionate influence by countries with relatively few funds, we weight observations by the number of funds in a given country in estimating Eq. (2), effectively equal-weighting each fund (as opposed to each country).[4] The primary variable of interest, $PostMMoU_{it}$, is an indicator that equals one if country i signed the MMoU before quarter t, zero if country i had not applied to sign the MMoU before quarter t, and missing if quarter t is between country i's application to sign the MMoU and the date of signing. We exclude the period between a country's MMoU application and signing dates (two years on average) because during this transition period, the laws to facilitate information sharing might be in place, but the administrative approvals might not yet be complete, making it unclear how investors will respond. We expect a positive $PostMMoU_{it}$ coefficient, indicating that funds tilt their investment portfolios toward treated (US-cross-listed) firms relative to control firms in the same country following the MMoU signing.

In each specification, we explicitly control for changes in the dependent variable that arise from shifts in the market capitalization of available US-cross-listed shares relative to the total investable shares in a country (e.g., because of new US cross-listings or because of changes in the market value of existing US cross-listings). This control variable, $\frac{XLMarketCap_{ijt}}{MarketCap_{ijt}}$, is calculated in a similar fashion as the dependent variable: the numerator is the aggregate market capitalization of all US-cross-listed firms in country i in quarter t, divided by the aggregate market

[3] Unless otherwise noted, we exclude firms cross-listed on the US Over-The-Counter (OTC) market because the level of SEC supervision of these firms is less clear than for exchange-listed or non-cross-listed firms. For ease of exposition, we refer to firms that are either cross-listed on a US exchange or are not cross-listed in the US at all as "all firms," and firms that cross-list on a US exchange as "cross-listed" firms. We exclude unsponsored ADRs from our analyses altogether because unsponsored listings are typically originated by a US bank without the involvement of the firm.

[4] For example, the US has about 43 times more sample funds than Portugal. Absent weighting, a given fund in Portugal would have 43 times more weight than one in the US. In additional (untabulated) analyses, we confirm that our inferences are unchanged if we instead: (1) use no weights and limit the sample to the top-30 fund countries to limit the influence of countries with relatively few funds, (2) weight by aggregate fund market value, (3) restrict the maximum weight of any single country to 1000 funds or less, (4) exclude any of the top-10 largest fund countries, or (5) use no regression weights, effectively equal-weighting the sample by fund country (however this latter result is slightly weaker than our baseline result).

capitalization of all firms in country i in quarter t. γ_{ij} is a time-invariant fixed effect for each fund-country/firm-country pair, included to control for any static factors that affect the level of investment between country pairs (e.g., a common currency). δ_{jt} is a time-varying fixed effect for each fund country and controls for the time-varying preferences of funds from a particular country for cross-listed stocks (in general). To account for correlation in the level of ownership over time, we cluster standard errors by firm country, fund country, and quarter.

3.3. Sample data

Ownership data are from FactSet, which provides disaggregated security-level holdings at the fund level for mutual funds from a wide range of countries. We use these data to construct fund-firm equity holdings by calendar quarter and aggregate these holdings as needed for each analysis. We exclude index or closed-end funds where managers' discretionary investment choices are constrained. For an investee country to be included in the sample, there must be at least ten cross-listed firms from that country, which leads to a final sample of funds from 65 investor countries in firms from 27 investee countries. Holdings data are from a 14-year window, beginning one year before the first country signs the MMoU (i.e., the sample spans from the 3rd quarter of 2001 through the 2nd quarter of 2015).

Table 1 Panel A reports the largest 30 of the 65 investor countries (the remaining 35 countries represent 1232 funds). About 22.8% of the funds are headquartered in the US with the remainder spread fairly evenly across a range of countries. All 27 investee countries included in the sample are listed in Table 1 Panel B, which reports the total amount of foreign investment in each country and the amount invested in cross-listed firms from that country. We designate any firm domiciled outside the US but with shares traded on a US exchange as "cross-listed." The US is excluded as an investee country because, by definition, US-cross-listed firms cannot be domiciled in the US. Among the investee countries, the UK attracts the most fund investment, but investment is spread across a wide range of countries. In a given target country, investment in cross-listed firms as a percentage of total investment (not controlling for other factors such as firm size) is typically above 50%, which is consistent with prior research indicating institutional investors' preference for US-cross-listed firms (Ferreira and Matos, 2008). Overall, the sample comprises about $1.6 trillion of total investment.

Table 1 Panel B also reports the date on which a country signed the MMoU, committing to cooperation with the SEC and other regulators. As is evident in the table, MMoU signature dates are spread over time with no clear divisions between emerging and developed markets. Rather, the variation in adoption timing is likely attributable to differences in existing laws across countries, which dictate the legislative process of enacting the new laws that are required to join the MMoU and the development of the required infrastructure to facilitate information sharing.

Table 2 reports descriptive statistics. The sample is fairly well-balanced across the pre- and post-MMoU pe-

Table 1
Sample composition.
The sample includes open-ended mutual fund holdings over a 14-year window (from the 3rd quarter of 2001 through the 2nd quarter of 2015) from FactSet. Panel A presents the number of mutual funds in the sample per fund country. The top 30 fund countries, measured by fund counts, are tabulated; the entire sample contains 46,372 funds across 65 countries (countries with fewer funds are included in the sample but not tabulated in Panel A for brevity). Panel B shows the investment in stocks (in billions of US dollars) in an investee country by sample funds, averaged by reporting quarter. For an investee country to be included in the sample, the country must have at least ten firms cross-listed in the US (in which funds are investing) in each quarter. *Cross-listed* investment refers to investment in companies that cross-list on a US exchange, whereas *All* investment includes companies that cross-list on a US exchange and those that do not. The MMoU signed date represents the date that country signed the International Organization of Securities Commissions' Multilateral Memorandum of Understanding.

Panel A: Mutual fund countries

Country	Funds	Country	Funds
United States	10,612	South Africa	707
Spain	5947	Belgium	597
France	4145	Denmark	570
United Kingdom	3405	Netherlands	499
Canada	3320	China	474
Japan	2135	Finland	465
Germany	1959	Taiwan	457
Austria	1106	Liechtenstein	440
India	1099	Norway	390
Italy	1051	Chile	299
Switzerland	953	Poland	280
Israel	878	Malaysia	271
Brazil	874	Luxembourg	270
Sweden	743	Thailand	250
Australia	715	Portugal	229

Panel B: Investee firm countries

Investee	Investment ($ Bil)		MMoU
country	*Cross-listed*	*All*	signed
United Kingdom	207.9	311.6	10 Mar 2003
Canada	173.3	259.6	23 Oct 2002
Japan	42.1	134.2	19 Feb 2008
France	53.8	107.7	19 Feb 2003
Switzerland	72.4	106.6	15 Feb 2010
Netherlands	63.6	77.7	22 Nov 2007
China	17.8	72.5	29 May 2007
India	9.2	56.4	22 Apr 2003
Germany	14.9	50.2	5 Nov 2003
South Korea	9.2	50.1	9 Jun 2010
Ireland	34.2	38.0	24 Dec 2012
Taiwan	14.1	37.5	15 Mar 2011
Brazil	26.3	37.2	21 Oct 2009
Spain	21.6	35.9	24 Mar 2003
Italy	12.3	34.1	15 Sep 2003
Sweden	9.0	33.1	17 May 2011
Australia	10.6	30.0	8 Oct 2002
Hong Kong	11.2	25.2	3 Mar 2003
Mexico	17.3	21.1	14 Mar 2003
Russia	0.7	20.8	16 Feb 2015
Bermuda	19.5	20.8	07 Jun 2007
South Africa	5.0	15.4	18 Mar 2003
Norway	3.8	12.7	11 Dec 2006
Singapore	4.0	12.1	17 Nov 2005
Israel	10.2	11.6	2 Jul 2006
Chile	1.7	2.9	–
Argentina	1.0	1.1	12 Jun 2014

Table 2
Descriptive statistics.
Panel A presents summary statistics for the variables of interest. Subscript i denotes investee firm country, j denotes mutual fund country, and t denotes time (year and quarter). All variables are defined in Appendix A. The sample includes open-ended mutual fund holdings over a 14-year window (from the 3rd quarter of 2001 through the 2nd quarter of 2015) from FactSet. Panel B presents the key partitioning variables used in the cross-sectional analyses. The sample for the OTC_l analysis is all funds, with potentially two observations per fund country, investee firm country, and quarter triple: one for exchange-listed cross-listings and one for sponsored OTC cross-listings (there are fewer observations for the OTC cross-listings because not all firm countries that have exchange-listed cross-listed firms also have OTC-cross-listed firms). The full sample is used for the $LocalFund_{ij}$ and $SECActivity_{ij}$ analyses, whereas non-local funds ($i \neq j$) are used for the $Contiguous_{ij}$, $SameLanguage_{ij}$, and $HighTrust_{ij}$ analyses (sample attrition for the $HighTrust_{ij}$ analysis is a result of the Eurobarometer survey only covering European countries). The sample for the $USFund_j$ partitioning uses only foreign fund observations ($i \neq j$), and the sample for the $NATO_j$ partitioning uses only non-US foreign fund observations ($j \neq$ US and $i \neq j$). The $Transient_s$ analysis uses observations from all fund countries, with potentially two observations per fund country, investee firm country, and quarter triple: one for transient funds and one for dedicated funds (not strictly doubled because not all fund countries have funds of both categories). The flows analysis uses all non-US, foreign funds ($j \neq$ US and $i \neq j$). Panel C presents pairwise correlations between select variables of interest. Spearman correlations are above the diagonal; Pearson correlations are below the diagonal. The Eurobarometer survey for $HighTrust_{ij}$ does not include US data, and hence no correlation is defined with $USFund_j$.

Panel A: Summary statistics

Variable	N	Mean	St. dev.	Pctl(25)	Median	Pctl(75)
Fund country × Investee firm country × Quarter variables:						
$\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100$	47676	53.476	33.072	25.735	54.537	83.264
$\frac{XLInvCount_{ijt}}{InvCount_{ijt}} \times 100$	47676	47.722	31.898	20.290	45.256	73.357
$\frac{Inv_{ijt}}{Inv_{jt}} \times 100$	47702	1.538	4.049	0.037	0.331	1.439
$\Delta\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100$	44998	−0.041	12.383	−1.550	0.000	1.222
Investee firm country × Quarter variables:						
$PostMMoU_{it}$	42503	0.683	0.465	0.000	1.000	1.000
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$	47676	50.329	25.530	29.802	53.588	71.286
$\frac{XLMarketCapFreeFloat_{it}}{MarketCapFreeFloat_{it}} \times 100$	45287	52.828	27.427	31.662	53.067	76.394
$\frac{XLFirms_{it}}{Firms_{it}} \times 100$	47676	13.801	18.041	2.111	5.237	20.192
$NIRatioVW_{it}$	47676	0.891	0.534	0.901	0.990	1.000
$DebtRatioVW_{it}$	47676	0.830	0.297	0.832	0.980	1.000
$LiqRatioVW_{it}$	47676	0.370	0.258	0.132	0.355	0.552
$NIRatioEW_{it}$	47676	0.745	1.402	0.692	0.908	0.995
$DebtRatioEW_{it}$	47676	0.720	0.311	0.501	0.863	0.979
$LiqRatioEW_{it}$	47676	0.083	0.142	0.007	0.016	0.098
$\frac{MarketCap_{it}}{MarketCap_t} \times 100$	47702	1.043	1.412	0.226	0.506	1.470
$\Delta\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$	44998	−0.042	2.965	−1.194	−0.004	1.039
Fund country × Quarter variables:						
$HiInflow_{jt}$	44998	0.254	0.435	0.000	0.000	1.000
$HiOutflow_{jt}$	44998	0.251	0.434	0.000	0.000	1.000
Fund country × Investee firm country variables:						
$LocalFund_{ij}$	47676	0.023	0.151	0.000	0.000	0.000
$Contiguous_{ij}$	46137	0.055	0.229	0.000	0.000	0.000
$SameLanguage_{ij}$	46137	0.167	0.373	0.000	0.000	0.000
$HighTrust_{ij}$	10163	0.378	0.485	0.000	0.000	1.000
Fund country variables:						
$USFund_j$	47676	0.034	0.181	0.000	0.000	0.000
$NATO_j$	47676	0.464	0.499	0.000	0.000	1.000
$USVotingSimilarity_j$	43389	0.209	14.528	−3.529	−0.245	2.036
Investee Firm country variables:						
$SECActivity_i$	47676	0.432	0.495	0.000	0.000	1.000
$USTradeImportance_i$	45877	−0.006	17.420	−11.555	−2.555	13.445

Panel B: Cross-sectional sample splits

Table	Sample Split		Observations	Mean value of: $\frac{XLInv_{ijtl}}{Inv_{ijtl}} \times 100$	$PostMMoU_{it}$	$\frac{XLMarketCap_{itl}}{MarketCap_{itl}} \times 100$
3-B	OTC_l	= 0	42503	53.913	0.683	50.915
		= 1	37536	1.221	0.694	1.850
				$\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100$	$PostMMoU_{it}$	$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$
4	$LocalFund_{ij}$	= 0	41501	54.351	0.682	50.933
		= 1	1002	35.755	0.716	50.169
4	$Contiguous_{ij}$	= 0	37779	55.161	0.682	51.328
		= 1	2350	41.953	0.656	44.082
4	$SameLanguage_{ij}$	= 0	33154	54.849	0.663	50.775
		= 1	6975	52.194	0.764	51.515
4	$HighTrust_{ij}$	= 0	5772	50.744	0.581	51.990
		= 1	2948	50.620	0.560	54.586

(*continued on next page*)

Table 2 (*continued*)

5	$SECActivity_i$	= 0	23900	48.613	0.600	43.843
		= 1	18603	60.723	0.789	60.001
6	$Transient_s$	= 0	41426	53.854	0.682	52.203
		= 1	28689	53.206	0.688	51.912
7-A	$USFund_j$	= 0	40042	54.173	0.686	50.911
		= 1	1459	59.252	0.565	51.518
7-B	$NATO_j$	= 0	22215	52.334	0.737	50.824
		= 1	17827	56.464	0.623	51.021
				$\Delta \frac{XLInv_{ijt}}{Inv_{ijt}} \times 100$	$PostMMoU_{it}$	$\Delta \frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$
8-B	$HiInflow_{ij}$	= 1	9746	−0.152	0.682	0.030
	$HiInflow_{ij} + HiOutflow_{ij}$	= 0	19262	−0.137	0.666	−0.214
	$HiOutflow_{ij}$	= 1	9758	0.132	0.770	0.046

Panel C: Correlation table

	Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1)	$\frac{XLInv_{ijt}}{Inv_{ijt}}$		−0.03	0.60	0.57	0.21	0.21	0.45	−0.08
(2)	$PostMMoU_{it}$	−0.02		0.02	0.03	0.03	0.05	−0.15	0.01
(3)	$\frac{XLMarketCap_{it}}{MarketCap_{it}}$	0.61	0.03		0.95	0.39	0.36	0.71	−0.00
(4)	$\frac{XLMarketCapFreeFloat_{it}}{MarketCapFreeFloat_{it}}$	0.58	0.04	0.95		0.39	0.33	0.67	0.00
(5)	$NIRatioVW_{it}$	0.14	0.05	0.22	0.22		0.76	0.27	0.00
(6)	$DebtRatioVW_{it}$	0.35	0.17	0.51	0.51	0.38		0.31	−0.01
(7)	$LiqRatioVW_{it}$	0.45	−0.19	0.70	0.66	0.15	0.37		−0.02
(8)	$LocalFund_{ij}$	−0.08	0.01	−0.00	0.00	−0.00	−0.01	−0.02	
(9)	$Contiguous_{ij}$	−0.09	−0.01	−0.06	−0.03	−0.02	−0.12	−0.06	−0.04
(10)	$SameLanguage_{ij}$	−0.02	0.08	0.01	−0.01	−0.02	0.02	0.01	−0.07
(11)	$HighTrust_{ij}$	−0.01	−0.01	0.06	0.09	0.02	−0.06	−0.05	0.28
(12)	$USFund_j$	0.03	−0.05	0.00	0.00	−0.00	−0.00	0.01	−0.03
(13)	$SECActivity_i$	0.19	0.20	0.33	0.41	0.08	0.09	0.36	0.01
(14)	$USTradeImportance_i$	0.14	−0.05	0.09	0.01	−0.04	−0.02	0.16	−0.03
(15)	$NATO_j$	0.07	−0.12	0.01	0.01	−0.01	0.00	0.03	−0.01
(16)	$USVotingSimilarity_j$	0.04	−0.11	0.02	0.01	−0.01	−0.00	0.03	−0.05

	Variable	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
(1)	$\frac{XLInv_{ijt}}{Inv_{ijt}}$	−0.09	−0.02	−0.02	0.03	0.18	0.18	0.07	0.02
(2)	$PostMMoU_{it}$	−0.01	0.08	−0.01	−0.05	0.20	−0.06	−0.12	−0.14
(3)	$\frac{XLMarketCap_{it}}{MarketCap_{it}}$	−0.06	0.01	0.04	0.00	0.32	0.12	0.01	0.01
(4)	$\frac{XLMarketCapFreeFloat_{it}}{MarketCapFreeFloat_{it}}$	−0.03	−0.01	0.06	0.00	0.42	0.04	0.01	0.01
(5)	$NIRatioVW_{it}$	−0.07	−0.03	0.02	−0.00	0.03	−0.09	−0.00	−0.00
(6)	$DebtRatioVW_{it}$	−0.13	0.03	−0.11	0.00	−0.07	−0.01	0.01	0.00
(7)	$LiqRatioVW_{it}$	−0.06	0.01	−0.05	0.01	0.41	0.17	0.03	0.03
(8)	$LocalFund_{ij}$	−0.04	−0.07	0.28	−0.03	0.01	−0.02	−0.01	−0.03
(9)	$Contiguous_{ij}$		0.16	0.13	0.02	0.10	−0.14	0.04	0.06
(10)	$SameLanguage_{ij}$	0.16		0.09	0.20	−0.05	0.13	−0.07	−0.04
(11)	$HighTrust_{ij}$	0.13	0.09			0.13	0.11	−0.31	−0.00
(12)	$USFund_j$	0.02	0.20			−0.01	0.01	0.20	0.33
(13)	$SECActivity_i$	0.10	−0.05	0.13	−0.01		−0.06	−0.01	−0.01
(14)	$USTradeImportance_i$	−0.14	0.14	0.06	0.01	−0.04		0.02	0.01
(15)	$NATO_j$	0.04	−0.07	−0.31	0.20	−0.01	0.01		0.70
(16)	$USVotingSimilarity_j$	0.03	0.12	−0.01	0.80	−0.01	0.00	0.45	

riods, with 68% of the observations in the post-MMoU period. The cross-sectional variables presented are on a country level (either firm country, fund country, or firm "country/fund-country pair) in Panel A. In Table 2 Panel B, for each cross-sectional split, we present the subsample size and means of the key variables in each subsample. Panel C tabulates the Spearman and Pearson correlation coefficient for each pair of variables.

4. Effect of the MMoU on foreign portfolio investment

Table 3 reports results for the effect of the MMoU on overall mutual-fund investment (i.e., including both foreign and local funds). The coefficient of interest, *PostMMoU*, captures fund-country *j*'s change in holdings of US-cross-listed stocks in country *i* (relative to non-cross-listed stocks in the same country) following the MMoU. Consistent with SEC oversight increasing institutional investor demand for US-cross-listed stocks, in Column 1, the coefficient estimate for $PostMMoU_{it}$ indicates that investors' tilt their portfolios (in terms of the relative dollar value of their investment) toward US-cross-listed firms after the investee country signs the MMoU. On average, fund managers increase their holdings of US-cross-listed firms by 6.7 percentage points relative to their total holdings in the investee country before the MMoU. The pre-MMoU mean of the dependent variable is 53 percentage points (i.e., 53 cents of each dollar of cross-border investment is allocated to US-cross-listed firms). The 6.7 percentage point increase implies a tilt toward US-cross-listed firms that is about 12.6 percent

Table 3
Effect of MMoU on fund investment.
Panel A presents estimates of regressions of:

$$y_{ijt} = \alpha_1 PostMMoU_{it} + \beta Controls_{it} + \gamma_{ij} + \delta_{jt} + \varepsilon_{ijt}.$$

Panel B presents estimates of regressions of:

$$y_{ijtl} = \alpha_1 PostMMoU_{it} + \alpha_2 (PostMMoU_{it} \times OTC_l) + \beta_1 Control_{itl} + \beta_2 (Control_{itl} \times OTC_l) + \gamma_{ijl} + \delta_{jtl} + \varepsilon_{ijtl}$$

where i indexes investee firm country, j indexes fund country, t indexes time (quarter and year), and l indexes cross-listing type (on a US exchange or sponsored OTC). $\gamma_{ij[l]}$ and $\delta_{jt[l]}$ are fixed effects. All variables are defined in Appendix A. The dependent variables are located in the column headings; control variables differ across the specifications as described in the table. The sample, detailed in Tables 1and 2, is open-ended mutual fund holdings from the 3rd quarter of 2001 through the 2nd quarter of 2015. Observations are weighted by the number of funds in country j. All models are estimated with observations weighted by the number of funds in country j. Standard errors (in parentheses) are clustered by firm country, fund country, and quarter. *, **, *** indicate two-sided p-values less than 10%, 5%, and 1%, respectively.

Panel A: Effect on firms cross-listed on US exchanges

	$\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100$			$\frac{XLInvCount_{ijt}}{InvCount_{ijt}} \times 100$	
	(1)	(2)	(3)	(4)	(5)
$PostMMoU_{it}$	6.680**	7.525**	6.150**	4.870*	4.226*
	(3.016)	(3.422)	(2.812)	(2.546)	(2.265)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$	0.654***		0.606***		
	(0.091)		(0.094)		
$\frac{XLMarketCapFreeFloat_{it}}{MarketCapFreeFloat_{it}} \times 100$		0.353***			
		(0.058)			
$NIRatioVW_{it}$			−0.543		
			(0.347)		
$DebtRatioVW_{it}$			7.571**		
			(3.164)		
$LiqRatioVW_{it}$			−0.996		
			(3.301)		
$\frac{XLFirms_{it}}{Firms_{it}} \times 100$				0.572***	0.808***
				(0.107)	(0.191)
$NIRatioEW_{it}$					0.199
					(0.451)
$DebtRatioEW_{it}$					11.068**
					(5.544)
$LiqRatioEW_{it}$					−24.236***
					(9.177)
Fixed effects:					
Fund country × Investee country	Yes	Yes	Yes	Yes	Yes
Fund country × Quarter	Yes	Yes	Yes	Yes	Yes
Standard error clusters:					
Fund country	65	64	65	65	65
Investee country	27	26	27	27	27
Quarter	60	60	60	60	60
N	42,503	39,225	42,503	42,503	42,503
R^2	0.797	0.782	0.798	0.869	0.872

Panel B: Comparison of cross-listing type

	$\frac{XLInv_{ijtl}}{Inv_{ijtl}} \times 100$	$\ln\left(1 + \frac{XLInv_{ijtl}}{Inv_{ijtl}} \times 100\right)$
	(1)	(2)
$PostMMoU_{it}$	6.680**	0.208**
	(2.989)	(0.083)
$PostMMoU_{it} \times OTC_l$	−6.694**	−0.201*
	(3.049)	(0.116)
$\frac{XLMarketCap_{itl}}{MarketCap_{itl}} \times 100$	0.654***	
	(0.090)	
$\frac{XLMarketCap_{itl}}{MarketCap_{itl}} \times 100 \times OTC_l$	−0.047	
	(0.102)	
$\ln\left(1 + \frac{XLMarketCap_{itl}}{MarketCap_{itl}} \times 100\right)$		0.910***
		(0.134)
$\ln\left(1 + \frac{XLMarketCap_{itl}}{MarketCap_{itl}} \times 100\right) \times OTC_l$		−0.226
		(0.150)
Fixed effects:		
Fund country × Investee country × OTC	Yes	Yes
Fund country × Quarter × OTC	Yes	Yes
Standard error clusters:		
Fund country	65	65
Investee country × Listing type	54	54
Quarter	60	60
N	80,039	80,039
R^2	0.924	0.956
$\alpha_1 + \alpha_2$	−0.025	0.007
	(0.432)	(0.073)

higher than in the pre-MMoU period. The aggregate investment for our sample is \$1.6 trillion; thus, the coefficient implies an aggregate, post-MMoU shift of about \$110 billion toward cross-listed firms.

Although the research design controls for any investee-country changes that affect both cross-listed and non-cross-listed firms similarly, if the two types of firms experience differential changes in a characteristic that affects the relative demand for shares around the MMoU-signing date, these differential changes could bias the estimation of the treatment effect of the MMoU. Our primary approach for addressing this concern, discussed further below, is to exploit cross-sectional variation in investor and investee characteristics that are likely to be differentially affected by changes in regulatory oversight. However, as a first step, we include several additional controls in Eq. (2) and assess the effect of including these variables on the estimated treatment effect.

For some firms, a substantial portion of their equity is closely held and unavailable to outside investors (Dahlquist et al., 2003). In our research design, closely held ownership could only be an issue if it changes for cross-listed relative to non-cross-listed stocks around the signing of the MMoU, which seems unlikely given that closely held ownership tends to be relatively sticky. Nonetheless, to ensure that this is not a concern, in Column 2, we adjust the market capitalization control variable to account for free float, and find that the estimated effect of the MMoU increases slightly (7.5 versus 6.7) and remains statistically significant. Given that this control has little effect on the estimated treatment effect, we use our original market capitalization control variable for all subsequent analyses.

Prior research also shows that firm characteristics such as profitability, capital structure, and liquidity affect the demand for firms' shares (e.g., Ferreira and Matos, 2008). Column 3 includes controls for value-weighted differences in profitability ($NIRatioVW_{it}$), leverage ($DebtRatioVW_{it}$), and liquidity ($LiqRatioVW_{it}$) between cross-listed and non-cross-listed stocks. Controlling for profitability, leverage, and liquidity, the magnitude of the $PostMMoU_{it}$ coefficient estimate is similar to that in Column 1 (6.2 versus 6.7), suggesting that these factors (or other unobservable characteristics correlated with these variables) do not explain the observed change in mutual-fund ownership. We do not include these additional control variables in the subsequent analyses because their inclusion reduces the sample size and increases the standard errors of our coefficient estimates but does not materially affect the magnitude of these estimates.[5]

Next, we consider post-MMoU changes in mutual fund investment in US-cross-listed firms along the extensive margin (i.e., whether funds increase the number of US-cross-listed stocks they hold). If, for example, greater SEC oversight increases investor confidence, investors should open new positions in cross-listed shares. Examining the extensive margin also provides comfort that our results are not sensitive to how we measure the percentage of ownership. The dependent variable is the number of US-cross-listed firms held by funds from a given investor country scaled by the total number of firms (cross-listed and non-cross-listed) in the investee country owned by funds from the investor country. We construct a time-varying control variable in analogous fashion to capture changes in the number of cross-listed and non-cross-listed firms.

The results, tabulated in Table 3 Column 4, yield similar inferences to those in Column 1. Following the signing of the MMoU, the number of US-cross-listed firms held by funds in the signatory country increases by 4.9 percentage points relative to non-US-cross-listed firms. Given that cross-listed firms make up 48% of funds' total holdings (see Table 2), the actual increase is about 10.2%. This result suggests that much of the observed increase in the tilt toward US-cross-listed firms is on the extensive margin. Results controlling for profitability, leverage, and liquidity changes around the MMoU in Column 5 are similar. Because this analysis uses the count of firms rather than the dollar value of investment, we equal-weight firms in our calculation of the control ratios.

In the primary analyses, we only include cross-listed firms that trade on a US stock exchange and thus are registered with the SEC (i.e., Level II and Level III ADRs) because the SEC's mandate over those firms is clearest. To further investigate the role of SEC oversight in explaining the increased preference for US-cross-listed stocks, Table 3 Panel B expands the sample to examine the changes in fund holdings of firms with sponsored ADRs that are traded in the US over-the-counter market ("OTC firms") following a country's signing of the MMoU. Prior literature shows that the effects of cross-listing are concentrated in exchange-listed firms, likely reflecting their more stringent filing requirements and greater scrutiny by the SEC (relative to OTC firms) (Karolyi, 1998; Miller, 1999; Doidge, 2004; Doidge et al., 2004; Hail and Leuz, 2006). Consequently, the incremental effect of the MMoU should be largest for exchange-listed firms.

We generate two observations for each firm-country/fund-country/quarter triple: the first is identical to that in our main analysis; the second replaces investment in exchange-cross-listed firms with sponsored OTC-cross-listed firms. We then estimate Eq. (2), fully interacting the model with an indicator for OTC firms.[6] The $PostMMoU_{it} \times OTC_l$ interaction term captures the

[5] While the decrease in sample size is not apparent in the tables, which include observations at the fund-country, firm-country, and quarter levels, including the control variables does reduce the number of individual firms that underlie these country-quarter-level averages. In an additional untabulated analysis, we confirm that all analyses are robust to the inclusion of these control variables, with the following exceptions: two of the estimated coefficients in the cross-sectional analyses are no longer statistically significant when we include the additional control variables (Table 4 Column 2 and Table 7 Panel B), although the coefficient magnitudes remain similar.

[6] By fully interacting the model (i.e., interacting all the variables and fixed effects), this approach is econometrically equivalent to splitting the sample on the partitioning variable of interest and estimating the model separately for each sample split. By estimating the pooled sample, however, we can directly conduct a statistical test for differences in the coefficients across the two subsamples. The estimations are not identical because we typically parameterize the interaction with the partitioning variable so that the coefficient on the interaction term can be interpreted as an incremental effect. However, we also tabulate the total effects, which are identical to estimating the model separately on each subsample.

incremental effect of the MMoU signing for OTC-cross-listed firms relative to exchange-cross-listed firms. Consistent with increased SEC oversight driving foreign investors' response to the MMoU, the MMoU effect is significantly weaker for OTC than for exchange-listed firms. In fact, the combined PostMMoU coefficient indicates that there was no effect of the MMoU on funds' investment in OTC-cross-listed firms.

Given that investment in OTC firms is significantly less than in exchange-based cross-listings (see Table 2), it is possible that the small magnitude of investment makes it difficult to detect a change for the OTC subsample. To mitigate this concern, we reestimate the model using a log transformation of the dependent and control variables so that we can interpret the effect on a percentage-change basis. Our results are robust to this alternate specification: the MMoU effect is significant for the exchange-cross-listings but not for the sponsored OTC firms, and the difference between the two is statistically significant.

Overall, the evidence presented in this section suggests that after a country signs the MMoU, mutual funds increase both the number of positions held and their total investment in US-cross-listings in the signatory country. Based on these revealed preferences, it appears that the increase in SEC oversight accompanying the MMoU increased the relative attractiveness of cross-listed shares, resulting in substantial shifts of investment within MMoU-signing countries.

4.1. Assessing identification assumptions

An important assumption underlying our difference-in-differences analysis is that the pre-MMoU-signing-period trends in investment are similar for cross-listed and non-cross-listed firms. The staggered MMoU adoption dates and inclusion of time $\times$ fund country fixed effects should control for general investment trends in US-cross-listed firms that are common to all countries (regardless of whether they are MMoU signatories). However, it is still possible that, for the MMoU-signing countries, there are different trends in investment (not shared by the control countries) for cross-listed compared to non-cross-listed stocks. To assess this possibility, we conduct our analysis in event-time, estimating a regression in which, except for the quarter prior to the investee-country's MMoU application date (which serves as the benchmark period), we include an indicator for each quarter relative to the investee-firm-country's MMoU-signing date. Specifically, we include 23 indicators, starting with one for observations where t is 12 or more quarters before country i's application date, one for observations where t is 11 quarters prior, etc., up to t being 12 quarters or more after country i's signatory date. We also interact the market capitalization control variable with the event-time indicators to allow for changes in the relative size of the ADR and non-ADR markets over time.

Fig. 1 plots the coefficient estimates of the event-time indicators and associated confidence intervals. If the observed tilt reflects a more general trend in preferences for US-cross-listed stocks, the estimated coefficients prior to the MMoU adoption date should be statistically significant. As illustrated in Fig. 1, none of the coefficients is statistically distinguishable from zero during the pre-MMoU period and there is no evidence of a differential trend between cross-listed and non-cross-listed stocks over time. While we cannot formally test the parallel-trends assumption, Fig. 1 provides no evidence of a violation. Rather, the first significant change in the tilt toward US-cross-listed stocks occurs after the signing of the MMoU. In the post-MMoU period, the coefficients on $PostMMoU_{it}$ are consistently statistically positive, suggesting that there was a significant shift in preferences toward US-cross-listed firms after the MMoU signing. Next, we conduct several cross-sectional analyses to further explain the determinants of the investment changes.

4.2. Variation in the effect of the MMoU based on investor information asymmetry

This section examines how the effect of the MMoU varies based on the extent to which investors are likely to value additional regulatory oversight. In particular, investors at a greater informational disadvantage (absent the enhanced oversight) are most likely to benefit from the MMoU. We focus on characteristics of the investor country, the investee country, the investor–investee country pair, and the investment strategy of the fund.

4.2.1. Foreign vs. local investors

First, we compare the effect of the MMoU on local funds to its effect on foreign funds. By its extraterritorial nature, the MMoU was designed to protect foreign investors. The literature on home bias indicates that geographically proximate investors have an information advantage relative to non-local investors in terms of expertise in and access to the local legal systems, familiarity with domestic institutions and culture, and proximity to monitor local firms. Accordingly, greater SEC oversight should be less important for local investors (Bernile et al., 2014; Bae et al., 2008). In fact, local mutual funds could be an important source of shares to accommodate the increase in foreign demand if the MMoU decreases the relative advantage of being a local investor.

Furthermore, Baik et al. (2010) provide evidence of informational advantages for local investors even within the US, which implies that the local information advantage comes from sources beyond publicly disclosed financial reporting information and is likely to persist even among relatively more transparent US-cross-listed firms. The additional oversight associated with the MMoU has the potential to reduce the information advantage of local investors by extending the reach of US securities law to issues such as insider trading, market manipulation, and financial reporting quality. As a result, we predict a larger positive response to the MMoU by foreign funds.

To examine this question, we create an indicator for whether a fund is domiciled in the same country (i.e., a local investor) as the investee firm ($LocalFund_{ij}$). Table 4 reports the results of interacting this indicator fully with the main model. In Column 1, the increase in holdings by foreign funds following the signing of the MMoU ($PostMMoU_{it}$) is positive and statistically significant (6.9%), while the incremental effect of the MMoU on local funds

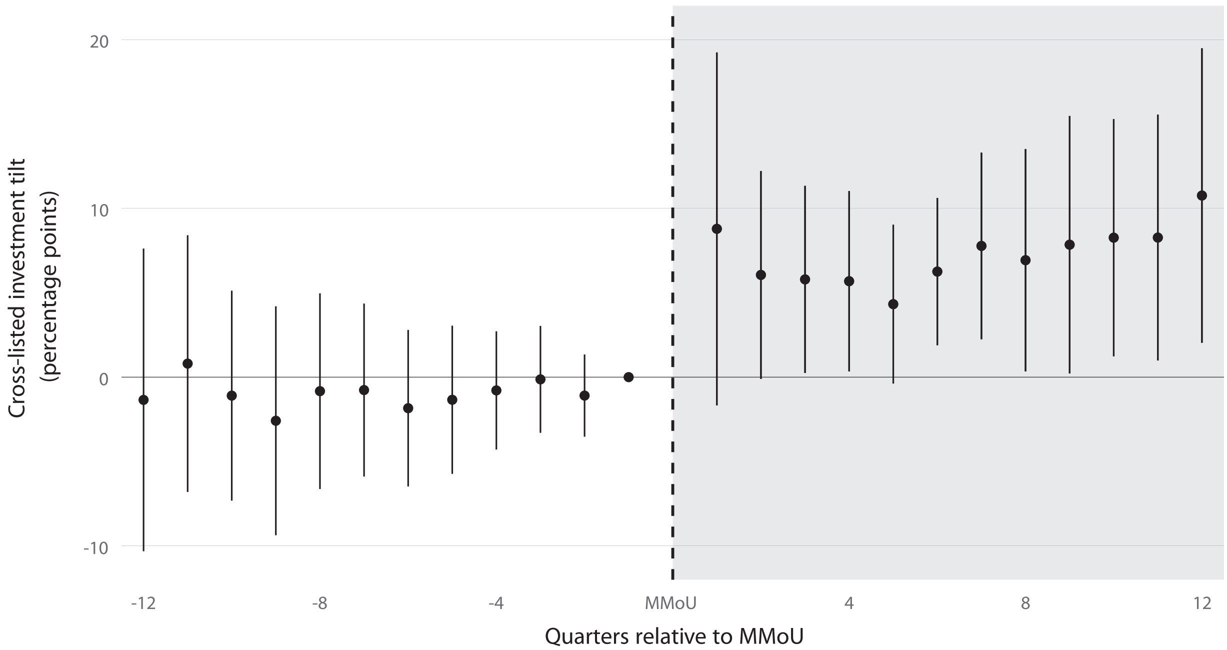

Fig. 1. Parallel trends event time analysis.
This figure plots the α coefficients and associated 95% confidence intervals from an estimation of the model:

$$\begin{aligned}\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100 = {} & \alpha_1 Minus12QandBeyond_{it} + \alpha_2 Minus11Q_{it} + \alpha_3 Minus9Q_{it} + \ldots \\ & + \alpha_{10} Minus3Q_{it} + \alpha_{11} Minus2Q_{it} + \alpha_{12} Plus1Q_{it} + \alpha_{13} Plus2Q_{it} + \ldots \\ & + \alpha_{22} Plus11Q_{it} + \alpha_{23} Plus12QandBeyond_{it} \\ & + \beta_1 \left(\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100 \times Minus12QandBeyond_{it} \right) + \ldots \\ & + \beta_{24} \left(\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100 \times Plus12QandBeyond_{it} \right) + \gamma_{ij} + \delta_{jt} + \varepsilon_{ijt}\end{aligned}$$

where i indexes firm country, j indexes fund country, and t indexes time (quarter and year). γ_{ij} is a time-invariant fixed effect for each fund-country firm-country pair, and δ_{jt} is a time-varying fixed effect for each fund country. $Minus12QandBeyond_{it}$ is an indicator if t is 12 or more quarters before country i applied to sign the MMoU; $Minus11Q_{it}$ is an indicator if t is 11 quarters before country i applied to sign the MMoU, and so on. $Plus1Q_{it}$ is an indicator if t is in the first quarter after country i signed the MMoU, and so on. The reference period is when t is in the quarter before country i's application to be a signatory to the MMoU. All other variables are defined in Appendix A. The sample includes open-ended mutual fund holdings over a 14-year window (from the 3rd quarter of 2001 through the 2nd quarter of 2015) from FactSet. Observations are weighted by the number of funds in country j. Standard errors are clustered by firm country, fund country, and quarter.

($PostMMoU_{it} \times LocalFund_{ij}$) is negative and significant (-10.2%), suggesting that local funds benefit less from the enhanced regulatory oversight accompanying the MMoU. While the sum of these coefficients is negative, the combined effect is not significantly different from zero. The absence of a positive effect for local funds also provides further assurance that our results do not reflect a general increase in the preference for US-cross-listed stocks by institutional investors.

4.2.2. Information asymmetry between the fund and firm country

The results in the previous section indicate that the adoption of the MMoU had a particularly pronounced effect for foreign investors, who are more likely to be at an information disadvantage relative to local investors. The extent of this informational disadvantage likely differs by specific investor-investee country pair. We proxy for the extent of information asymmetry and for the costs of monitoring to foreign investors based on whether the investor-investee country pair: (1) is not geographically contiguous, (2) does not share a common language, and (3) is characterized by a high level of distrust. For example, sharing the same language makes it easier to monitor a foreign firm as the direct cost of accessing and processing information is likely lower. Similarly, such countries are more likely to share cultural norms and expectations. Investors from countries separated by a lack of trust, as measured by the Eurobarometer trust indices (Guiso et al., 2009), are likely to place additional importance on the external oversight provided by the MMoU. For instance, Carlin et al. (2009) model the relation between regulation and trust in financial markets and show that the two act as substitutes. Consistent with this, Pinotti (2008) and Aghion et al. (2010) find that individuals prefer state control when trust is low.

Results reported in Table 4 Columns 2 through 4 support the prediction that the effect of the MMoU is largest

Table 4
Fund proximity.
Following are estimates of regressions of:

$$\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100 = \alpha_1 PostMMoU_{it} + \alpha_2 (PostMMoU_{it} \times Closeness_{ij}) + \beta_1 \left(\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100\right) + \beta_2 \left(\frac{XLMarketCap_{it}}{MarketCap_{it}} \times Closeness_{ij} \times 100\right) + \gamma_{ij} + \delta_{jtc} + \varepsilon_{ijt}$$

where i indexes investee firm country, j indexes fund country, t indexes time (quarter and year), and c indexes the *Closeness* measure. γ_{ij} is a time-invariant fixed effect for each fund country × investee firm country pair, and δ_{jt} is a time-varying fixed effect for each fund country × *Closeness*-measure-pair. In Column 1 the closeness measure is $LocalFund_{ij}$, in Column 2 $Contiguous_{ij}$, in Column 3 $SameLanguage_{ij}$, and in Column 4 $HighTrust_{ij}$. All variables are defined in Appendix A. The models are estimated with observations weighted by the number of funds in country *j*. The sample, detailed in Tables 1 and 2, is open-ended mutual fund holdings from the 3rd quarter of 2001 through the 2nd quarter of 2015; Columns 2 – 4 exclude observations where $i = j$. Standard errors (in parentheses) are clustered by firm country, fund country, and quarter. *, **, *** indicate two-sided *p*-values less than 10%, 5%, and 1%, respectively.

Dependent variable:	$\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100$			
Closeness variable:	$LocalFund_{ij}$ (1)	$Contiguous_{ij}$ (2)	$SameLanguage_{ij}$ (3)	$HighTrust_{ij}$ (4)
$PostMMoU_{it}$	6.852**	7.019**	8.495**	18.873***
	(3.058)	(3.299)	(3.417)	(4.556)
$PostMMoU_{it} \times Closeness_{ij}$	−10.242*	−9.278*	−5.224**	−20.231**
	(5.435)	(5.452)	(2.242)	(8.819)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$	0.651***	0.655***	0.610***	0.681***
	(0.091)	(0.097)	(0.101)	(0.085)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times Closeness_{ij} \times 100$	0.079	−0.008	0.167	0.160
	(0.156)	(0.213)	(0.117)	(0.207)
Fixed Effects:				
Fund country × Investee country	Yes	Yes	Yes	Yes
Fund country × Closeness × Quarter	Yes	Yes	Yes	Yes
Standard error clusters:				
Fund country	65	63	63	15
Investee country	27	28	28	12
Quarter	60	60	60	60
Sample funds	Full	Foreign	Foreign	Foreign
N	42,503	40,129	40,129	8,720
R^2	0.798	0.800	0.804	0.850
$\alpha_1 + \alpha_2$	−3.390	−2.259	3.270**	−1.358
	(3.058)	(4.482)	(1.452)	(5.886)

for investor-investee pairs where information asymmetries are likely the largest. The incremental effect for fund countries that are $Contiguous_{ij}$ to the MMoU-signing country is -9.278 and is statistically significant, suggesting that the effect of the MMoU is more pronounced when the investor and target countries are geographically separated. Similarly, the difference in coefficients for the incremental effect on funds from countries that share a $SameLanguage_{ij}$, reported in Column 3, is -5.224 and is statistically significant. Finally, the incremental effect for investor countries that have $HighTrust_{ij}$ of the investee country is -20.231 and statistically significant. Overall, the evidence suggests that the benefit of regulatory oversight is highest for funds that face higher costs to screen and monitor their investments.

4.2.3. Investee-country ties to the US

The incremental SEC oversight associated with the MMoU is also likely to vary based on the investee country's pre-MMoU ties to the US in general and the SEC in particular. In instances where the SEC previously experienced fewer barriers to cross-border enforcement, we expect the MMoU to have a more modest effect. Specifically, changes in foreign ownership will likely be smaller for signatory countries where the SEC had preexisting cooperation arrangements and a history of enforcement activity, as well as for countries with strong economic ties to the US prior to the MMoU.

To test this prediction, we first estimate the incremental effect of the MMoU in countries where the SEC had a bilateral arrangement and had pursued at least one enforcement action prior to the MMoU signing. We then construct a second, more general measure of the investee country's connection to the US, $USTradeImportance_i$, based on the importance of the US as a trading partner. Countries with a stronger trade connection are likely to have tighter preexisting economic and regulatory links (e.g., tax and trade policies and financial and security cooperation), making cross-border monitoring and enforcement less difficult. As a result, the MMoU should be less of a shock to enforcement. To calculate $USTradeImportance_i$, we use export data from the IMF's Direction of Trade Statistics database for the year 2000. To control for the size of the overall economy, for each country *i*, we scale exports by the destination country's Gross Domestic Product (GDP). We then percentile rank export destinations for each country *i* by scaled exports, with the most important partner ranked 99. The measure of US trade importance for country *i* is the percentile rank of the US as an export destination for that

Table 5
Investee country ties to the US.
This table presents estimates of regressions of:

$$\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100 = \alpha_1 PostMMoU_{it} + \alpha_2 (PostMMoU_{it} \times USTie_i) + \beta_1\left(\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100\right) + \beta_2\left(\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100 \times USTie_i\right) + \gamma_{ij} + \delta_{jt[c]} + \varepsilon_{ijt}$$

where i indexes firm country, j indexes fund country, t indexes time (quarter and year), and c indexes whether or not country i had previous SEC Activity. γ_{ij} is a time-invariant fixed effect for each fund country-firm country pair, and $\delta_{jt[c]}$ is a time-varying fixed effect for each fund country (times whether or not $SECActivity_i = 1$ in Column 1). The $USTie_i$ variable in Column 1 is $SECActivity_i$ (binary variable), whereas the variable in Column 2 is $USTradeImportance_i$ (continuous variable). All variables are defined in Appendix A. The sample, detailed in Tables 1 and 2, is open-ended mutual fund holdings from the 3rd quarter of 2001 through the 2nd quarter of 2015. Observations are weighted by the number of funds in country j. Standard errors are clustered by firm country, fund country, and quarter. *, **, *** indicate two-sided p-values less than 10%, 5%, and 1%, respectively.

Dependent variable:	$\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100$	
$USTie_i$ variable:	$SECActivity_i$ (1)	$USTradeImportance_i$ (2)
$PostMMoU_{it}$	10.285***	7.316**
	(3.723)	(3.073)
$PostMMoU_{it} \times USTie_i$	−11.561**	−0.171***
	(4.638)	(0.059)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$	0.664***	0.633***
	(0.125)	(0.089)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times USTie_i \times 100$	0.063	−0.005
	(0.148)	(0.004)
Fixed Effects:		
Fund country × Investee country	Yes	Yes
Fund country × Quarter	No	Yes
Fund country × Quarter × SECActivity	Yes	No
Standard error clusters:		
Fund country	65	65
Investee country	27	26
Quarter	60	60
N	42,503	41,503
R^2	0.808	0.804
$\alpha_1 + \alpha_2$	−1.275	
	(2.923)	

country. For ease of interpretation, we recenter this ranking so that the mean observation equals zero.

Table 5 reports results. In Column 1, consistent with our prediction, the coefficient estimate for the incremental effect of the MMoU in countries with stronger preexisting SEC enforcement links is negative and statistically significant (i.e., the $PostMMoU_{it} \times USTie_i$ coefficient is -11.561). Similarly, the results in Column 2 indicate that the effect of the MMoU is significantly smaller for signatories with stronger preexisting economic ties to the US (i.e., the $PostMMoU_{it} \times USTie_i$ coefficient is -0.171). In economic terms, the coefficient estimate indicates that a jump of 25 percentile points in the importance of the US as a trading partner leads to a 4.3 percentage point decrease in the magnitude of the investment shift associated with the MMoU.

Overall, these results provide evidence consistent with our prediction that the benefits of the MMoU are most pronounced for countries that were historically most distant from the US regulatory apparatus. The results also support the conclusion that our primary results reflect the effect of SEC oversight.

4.2.4. Fund strategy

It is also likely that there is variation in the importance of oversight (and thus the response to the MMoU) based on a fund's investment strategy. Following papers such as Bushee (1998, 2001) and Borochin and Yang (2017), we split investors into two groups (denoted by s) based on trading behavior: (1) transient investors and (2) dedicated investors using fund classifications from FactSet.[7] In general, transient investors trade frequently and hold a large number of smaller positions, while dedicated investors trade less frequently and hold a smaller number of larger positions. Borochin and Yang (2017) find that dedicated institutions tend to place a higher value on transparency and oversight because they take fewer positions, investing based on firm-specific fundamental information with a focus on holding the stock for long-term growth. Transient funds are more likely to be technical investors who trade based on short-term price movements and ar-

[7] Borochin and Yang (2017) split funds into Dedicated, Transient, and Quasi-Indexers using classifications from Bushee (1998, 2001). We cannot use the Bushee (1998, 2001) classifications directly because the methodology is based on SEC 13-F filings, which are only available for US funds.

Table 6
Fund strategy.
This table presents the estimate of a regression of:

$$\frac{XLInv_{ijts}}{Inv_{ijts}} \times 100 = \alpha_1 PostMMoU_{it} + \alpha_2 (PostMMoU_{it} \times Transient_s) + \beta_1 \left(\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100\right) + \beta_2 \left(\frac{XLMarketCap_{it}}{MarketCap_{it}} \times Transient_s \times 100\right) + \gamma_{ijs} + \delta_{jts} + \varepsilon_{ijts}$$

where i indexes firm country, j indexes fund country, t indexes time (quarter and year), and s indexes the fund strategy (transient or dedicated). γ_{ijs} is a time-invariant fixed effect for each fund country-firm country-fund strategy triple, and δ_{jts} is a time-varying fixed effect for each fund country times the fund strategy. The dependent variable, $\frac{XLInv_{ijts}}{Inv_{ijts}} \times 100$, is calculated for each quarter t as aggregate investment by funds employing strategy s from country j in firms from country i that cross-list on a US exchange divided by aggregate investment by funds employing strategy s from country j in firms from country i that either cross-list on a US exchange or do not cross-list in the US, times 100. All other variables are defined in Appendix A. The sample, described in detail in Tables 1 and 2, is all funds aggregated to the fund country × investee country × quarter × fund strategy level. Observations are weighted by the number of funds in country j. Standard errors are clustered by firm country, fund country, and quarter. *, **, *** indicate two-sided p-values less than 10%, 5%, and 1%, respectively.

	$\frac{XLInv_{sijt}}{Inv_{sijt}} \times 100$
$PostMMoU_{it}$	6.004**
	(2.574)
$PostMMoU_{it} \times Transient_s$	−2.321***
	(0.862)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$	0.692***
	(0.065)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times Transient_s \times 100$	−0.005
	(0.054)
Fixed effects:	
Fund country × Investee country × Transient	Yes
Fund country × Quarter × Transient	Yes
Standard error clusters:	
Fund country	65
Investee country	27
Quarter	60
N	70,115
R^2	0.740
$\alpha_1 + \alpha_2$	3.684
	(2.753)

bitrage. As a result, we expect dedicated funds to respond more to the MMoU because they tend to be fundamentals-based investors who value strong governance facilitated by better oversight.

Similar to Table 4, we create separate measures of transient and dedicated funds' relative preferences for US-cross-listed firms for each fund-country/firm-country/quarter triple. We then estimate Eq. (2), allowing each coefficient to vary by the type of investor (i.e., we fully interact the model with an indicator for $Transient_s$ funds). The results in Table 6 provide support for the prediction that the MMoU has a greater impact on dedicated investors. Dedicated investors make a significant shift towards US-cross-listed firms after the MMoU signing (i.e., the coefficient on $PostMMoU_{it}$ is 6.004), whereas the incremental effect for transient investors is significantly negative (i.e., the coefficient on $PostMMoU_{it} \times Transient_s$ is -2.321) and the total effect (i.e., $\alpha_1 + \alpha_2$) is not significantly different from zero. Overall, these results suggest that fundamentals-based, long-term investors who value strong governance benefit most from the enhanced regulatory oversight accompanying the MMoU.

4.3. Variation in the effect of the MMoU between US and non-US foreign investors

While the SEC's primary mandate is to protect US investors, there are reasons to expect that the benefits of the MMoU could be larger for non-US foreign investors. Prior literature suggests that US institutional investors have an information advantage over other foreign institutions because their substantial research capacity and location permit them to exploit a richer set of global, private information (e.g., Albuquerque et al., 2009; Ferreira et al., 2017). In addition, by virtue of their location, resources, and scale, US institutions are likely able to better assert their private legal right of action and governance preferences, even absent the MMoU (Fang et al., 2015). To the extent that non-US investors are a primary beneficiary of the stronger SEC oversight accompanying the MMoU, it suggests a (presumably unintended) regulatory spillover.

To investigate whether there is a spillover effect of the MMoU to non-US investors, we separately examine changes in investment for non-US and US (foreign) investors. Table 7 Panel A reports the results of this analysis. Column 1 shows the results for US investors relative to non-US-foreign investors when the specification is fully interacted with the $USFund_j$ indicator. Consistent with the benefit of SEC oversight extending beyond US investors, the positive coefficient estimate of 7.538 suggests that *non-US* foreign investors significantly increase their holdings of US-cross-listed firms after a country signs the MMoU. In fact, the significant negative coefficient estimate of -2.410 suggests that US investors actually benefit less from the MMoU than other foreign investors, although the overall effect for US investors, $\alpha_1 + \alpha_2$, is positive and statistically significant (5.127).

Furthermore, among non-US foreign investors, the MMoU should have a larger impact on funds from countries that have greater familiarity with and trust in US oversight. We use two measures of non-US foreign investors' alignment with the US: (1) an indicator for whether the fund country is a member of the NATO, and (2) the similarity between the investor country and the US in UN General Assembly voting. Both measures are intended to capture the policy and political alignment between the US and the investor country. NATO countries and countries that strongly align with the US in UN voting should have similar political structures and are more likely to have confidence in US government institutions. We measure similarity in UN voting using the percentile rank of the number of congruent votes between the fund country and the US from 1991 to 2000, with higher values indicating more agreement. This type of confidence is likely to be relatively static. To capture this time-invariant characteristic, we select the decade just prior to the start of our sample period. For ease of interpretation, we recenter the sample mean of the percentile ranks to zero.

Table 7 Panel B reports results. Column 1 indicates that the effect of the MMoU is significantly stronger for funds from NATO countries. The economic magnitude is

Table 7
Regulatory spillovers to non-US, foreign investors.
This table presents estimates of:

$$\begin{aligned}\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100 = {} & \alpha_1 PostMMoU_{it} \\ & + \alpha_2 (PostMMoU_{it} \times Cross\text{-}sectional_j) \\ & + \beta_1 \left(\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100\right) \\ & + \beta_2 \left(\frac{XLMarketCap_{it}}{MarketCap_{it}} \times Cross\text{-}sectional_j \times 100\right) \\ & + \gamma_{ij} + \delta_{jt} + \varepsilon_{ijt}\end{aligned}$$

where i indexes investee country, j indexes fund country, and t indexes time (quarter and year). γ_{ij} is a time-invariant fixed effect for each fund country × investee country pair, and δ_{jt} is a time-varying fixed effect for each fund country. All variables are defined in Appendix A. The samples, described in detail in Tables 1 and 2, are all foreign funds ($i \neq j$) in Panel A and all non-US foreign funds in Panel B. Observations are weighted by the number of funds in country j. Standard errors are clustered by firm country, fund country, and quarter. *, **, *** indicate two-sided p-values less than 10%, 5%, and 1%, respectively.

Panel A: US and non-US foreign investors

	$\frac{XLInv_{ijt}}{Inv_{ijt}} \times 100$
$PostMMoU_{it}$	7.538**
	(3.112)
$PostMMoU_{it} \times \text{USFund}_j$	−2.410***
	(0.017)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$	0.661***
	(0.096)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times USFund_j \times 100$	−0.028***
	(0.010)
Fixed effects:	
Fund country × Investee country	Yes
Fund country × Quarter	Yes
Standard error clusters:	
Fund country	65
Investee country	27
Quarter	60
N	41,501
R^2	0.796
$\alpha_1 + \alpha_2$	5.127**
	(2.138)

Panel B: Fund countries with close US links

Dependent variable:	$\frac{X(LInv_{ijt}}{Inv_{ijt}} \times 100$	
USLink variable:	$NATO_j$ (1)	$USVotingSim_j$ (2)
$PostMMoU_{it}$	5.468*	6.932**
	(3.147)	(2.956)
$PostMMoU_{it} \times \text{USLink}_j$	2.735*	0.222*
	(1.602)	(0.117)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$	0.578***	0.668***
	(0.086)	(0.094)
$\frac{XLMarketCap_{it}}{MarketCap_{it}} \times USLink_j \times 100$	0.106**	−0.002
	(0.053)	(0.004)
Fixed effects:		
Fund country × Investee country	Yes	Yes
Fund country × Quarter	Yes	Yes
Standard error clusters:		
Fund country	64	57
Investee country	27	27
Quarter	60	60
N	40,042	36,334
R^2	0.763	0.760
$\alpha_1 + \alpha_2$	8.203***	
	(3.119)	

substantial, with an increase from 5.5 percentage points for funds from non-NATO countries to 8.2 for funds from NATO countries. In Column 2, the coefficient capturing the interaction between UN voting and MMoU signing is positive and statistically significant. In terms of economic magnitude, following the MMoU signing, funds from a country voting with the US at the third quartile increased their portfolio tilt toward US-cross-listed firms by about 1.2 percentage points more than did funds from the first-quartile voting-similarity rank.

Overall, the results in this section suggest that there are significant spillover effects of the MMoU to non-US investors. Combined with the cross-sectional results from prior sections, the evidence suggests there is significant heterogeneity in investors' responses to the MMoU. Specifically, we find that the extent to which funds increase their holdings of US-cross-listed firms is associated with: (1) the level of information asymmetry between the investor and investee countries, (2) the investee country's preexisting ties to the US, and (3) the investor country's political alignment with the US. These results all point toward increased SEC oversight as an important mechanism through which the MMoU affects foreign portfolio investment. However, we caution that because these results are cross-sectional in nature, they are subject to endogeneity concerns (e.g., omitted variables correlated with the partitioning variables) and should be interpreted accordingly.

4.4. Post-MMoU portfolio reallocation approach

Finally, we provide descriptive evidence on how institutions adjust their portfolios to reach their desired post-MMoU portfolio allocations. Although our results suggest that foreign investors increase their holdings in US-cross-listed firms after a country signs the MMoU, they do not speak to how funds rebalance their portfolios to achieve this reallocation. In addition, exploring the costs that funds are willing to bear to achieve their desired post-MMoU portfolio allocations provides evidence on the economic significance of our results. We expect funds to start with the lowest cost method and move progressively toward more costly approaches. There are three notable possibilities, ranked from the highest to the lowest in terms of likely costs: (1) reallocating investments across countries (i.e., from non-MMoU countries to MMoU countries); (2) reallocating investments within a country (i.e., from non-ADR to ADR shares); and (3) disproportionately redirecting net fund flows to US-cross-listed shares in MMoU-signing countries.

Shifting investment across countries is likely the most costly reallocation approach because in addition to direct transactions costs, it also affects funds' target-country-level exposures and many funds likely have target levels of exposure that limit their ability to easily change allocations across countries. Shifting allocations within-country avoids changes to country-level exposures, but creates transactions costs on both the purchase and sale. Redirecting fund flows is cheapest because there need not be any incremental transactions costs (i.e., fund flows necessitate purchases or sales) to achieve the desired allocation.

Table 8
Post-MMoU reallocation approach.
Panel A Column 1 presents the estimate of the regression:

$$\frac{Inv_{ijt}}{Inv_{jt}} \times 100 = \alpha_1 PostMMoU_{it} + \beta_1\left(\frac{MarketCap_{it}}{MarketCap_t} \times 100\right) + \gamma_{ij} + \delta_{jt} + \varepsilon_{ijt}$$

whereas Column 2 presents the estimate of the regression:

$$\frac{Inv_{ijtl}}{Inv_{jt}} \times 100 = \alpha_1 (PostMMoU_{it} \times XL_l) + \beta_1\left(\frac{MarketCap_{itl}}{MarketCap_t} \times 100\right) + \gamma_{ijl} + \delta_{it} + \lambda_{tl} + \varepsilon_{ijtl}$$

where i indexes investee country, j indexes fund country, t indexes time (quarter and year), and l indexes listing type (cross-listed or not cross-listed). In Column 1, observations are aggregated to the fund country × investee country × quarter level; in Column 2, there are double the number of observations: one for each fund country × investee country × quarter × listing type (cross-listed or not). In Column 1, γ_{ij} is a time-invariant investee country × fund country fixed effect; δ_{jt} is a time-varying fixed effect for the fund country. In Column 2, γ_{lij} is a time-invariant fixed effect for a fund country × investee country × listing type triple, δ_{it} is a time-varying fixed effect for investee country i, and λ_{lt} is a time-varying fixed effect for listing type l. All variables are defined in Appendix A. In Column 1, standard errors (in parentheses) are clustered by fund country, firm country, and quarter; in Column 2 they are clustered by fund country, firm country × listing type, and quarter. Panel B presents estimates of regressions of:

$$\begin{aligned}\Delta \frac{XLInv_{ijt}}{Inv_{ijt}} \times 100 = {} & [\alpha_1 HiInflow_{jt} + \alpha_2 HiOutflow_{jt} + \alpha_3 PostMMoU_{it} \; +] \\ & \alpha_4 (HiInflow_{jt} \times PostMMoU_{it}) + \alpha_5 (HiOutflow_{jt} \times PostMMoU_{it}) \\ & + \left[\alpha_6 \left(\Delta \frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100\right) + \right] \gamma_{ij} + \{\delta_{jt} + \lambda_{it}\} + \varepsilon_{ijt}.\end{aligned}$$

Terms in square brackets are included in Column 1, but the inclusion of additional fixed effects in Column 2 in curly brackets subsumes the bracketed terms. i subscripts investee country, j subscripts fund country, and t subscripts quarter t. Δ denotes the difference from the prior quarter. γ_{ij} is a time-invariant fixed effect for the investee country × firm country pair, δ_{jt} is a time-varying fund country fixed effect, and λ_{it} is a time-varying investee country fixed effect. Standard errors (in parentheses) are clustered by fund country, firm country, and quarter. The sample, described in detail in Tables 1 and 2, is all non-US, foreign firms. *, **, *** indicate two-sided p-values less than 10%, 5%, and 1%, respectively.

Panel A: Worldwide allocation

	$\frac{Inv_{ijt}}{Inv_{jt}} \times 100$ (1)	$\frac{Inv_{lijt}}{Inv_{jt}} \times 100$ (2)
$PostMMoU_{it}$	−0.182	
	(0.172)	
$PostMMOU_{it} \times XL_l$		0.233**
		(0.111)
$\frac{MarketCap_{it}}{MarketCap_t} \times 100$	0.626***	
	(0.152)	
$\frac{MarketCap_{itl}}{MarketCap_t} \times 100$		0.673***
		(0.153)
Fixed effects:		
Investee country × Fund country	Yes	No
Fund country × Quarter	Yes	No
Listing type × Investee country × Fund country	No	Yes
Investee country × Quarter	No	Yes
Listing type × Quarter	No	Yes
Standard error clusters:		
Fund country	64	64
Investee country	27	
Investee country × Listing type		54
Quarter	60	60
N	41,143	82,286
R^2	0.840	0.768

Panel B: Fund flows

	$\Delta \frac{XLInv_{ijt}}{Inv_{ijt}} \times 100$	
	(1)	(2)
$HiInflow_{jt}$	−0.165	
	(0.213)	
$HiOutflow_{jt}$	−0.065	
	(0.298)	
$PostMMOU_{it}$	−0.244	
	(0.270)	
$HiInflow_{jt} \times PostMMOU_{it}$	0.518*	0.536
	(0.281)	(0.378)
$HiOutflow_{jt} \times PostMMOU_{it}$	0.906**	0.763**
	(0.409)	(0.338)
$\Delta \frac{XLMarketCap_{it}}{MarketCap_{it}} \times 100$	0.309***	
	(0.082)	
Fixed effects:		
Investee country × Fund country	Yes	Yes
Investee country × Quarter	No	Yes
Fund country × Quarter	No	Yes
Standard error clusters:		
Fund country	64	64
Investee country	27	27
Quarter	59	59
N	38,766	38,766
R^2	0.025	0.210

In our primary analyses, we cannot disentangle cross-country from within-country investment shifts because the denominator of the dependent variable is the total investment (by a given fund country) in a single investee country. Thus, to separate cross-country from within-country investment shifts, we construct an alternative measure of investment, $\frac{Inv_{ijt}}{Inv_{jt}}$, that uses a fund-country's investment in all firms in a given MMoU-signing country as the numerator and the fund-country's total worldwide investment as the denominator. This variable allows us to examine whether the MMoU leads to changes in funds' cross-country portfolio allocations. However, given that we can no longer use non-cross-listed firms as a control group, this analysis is prone to endogeneity concerns arising from other contemporaneous, investee-country-level changes. Furthermore, an implicit assumption in this analysis is that funds' investment flows are not materially affected by the MMoU (i.e., investors did not reallocate across funds based on the funds' exposure to cross-listed firms in MMoU-adopting countries).

Table 8 Panel A Column 1 reports results based on total investment by a given fund-country in an MMoU-signing country as a percentage of the fund-country's total worldwide investment. The insignificant coefficient on the MMoU indicator suggests that the MMoU was not associated with a significant shift in funds' cross-country allocations. Column 2 includes a separate measure of global investment for cross-listed and non-cross-listed firms (listing type denoted by l). This measure is similar to the main analysis in Table 3, with the exception that we measure funds' holdings in cross-listed firms as a fraction of total worldwide investment. Consistent with our prior results, the coefficient on $PostMMoU_{it} \times XL_l$ is positive and statistically significant, indicating that following the MMoU, mutual funds increased their holdings of US-cross-listed firms by 0.233 percentage points of total worldwide investment.

Next, we examine how funds allocate their net fund flows (i.e., the increases or decreases in a fund's assets under management that are not driven by price changes of existing investments) within a country following the signing of the MMoU. The incremental transactions costs should be lower when a fund needs to buy or sell shares because of new fund flows. In this situation, the firm can simply redirect those flows to change the tilt between cross-listed and non-cross-listed stocks following the MMoU signing, while avoiding any incremental transactions costs. Table 8 Panel B Column 1 estimates a regression of the model:

$$\begin{aligned}\Delta\frac{XLInv_{ijt}}{Inv_{ijt}}\times 100 = {} & \alpha_1 HiInflow_{jt} \\ & +\alpha_2 HiOutflow_{jt} \\ & +\alpha_3 PostMMoU_{it} \\ & +\alpha_4 (HiInflow_{jt}\times PostMMoU_{it}) \\ & +\alpha_5 (HiOutflow_{jt}\times PostMMoU_{it}) \\ & +\alpha_6 (\Delta\frac{XLMarketCap_{it}}{MarketCap_{it}}\times 100) \\ & +\gamma_{ij}+\varepsilon_{ijt} \qquad (3)\end{aligned}$$

where the dependent variable captures changes in funds' tilts toward US cross-listings in a given investee country from quarter t to $t-1$. $HiInflow_{jt}$ ($HiOutflow_{jt}$) is an indicator equal to one if fund-country j's fund flows in quarter t are in the top (bottom) quartile of the sample. We measure net fund flows by aggregating each fund's quarterly change in holdings arising from increases or decreases in assets under management that are not driven by price changes in existing investments. This approach assumes that all inflows (outflows) of capital to the fund are immediately used to increase or decrease equity holdings. We measure flows as a percentage of lagged total equity investment by the fund country, using this scaled measure to determine into which quartile of distribution the fund-country/quarter-year observation falls. The main effects (α_1 and α_2) indicate whether a fund-country's allocation towards cross-listings shifts as a function of net inflows or outflows. The interaction coefficients (α_3 and α_4) capture whether this relation changes after the MMoU is signed.

Table 8 Panel B reports results. The coefficients for the main effects of fund flows on investment are insignificant. However, the interactions are statistically significant, suggesting that in the post-period, funds take advantage of fund flows to reallocate their investment toward cross-listings. The positive coefficients indicate that, following an investee country signing the MMoU, funds with large inflows are more likely to buy cross-listed stocks, while funds with large outflows are more likely to sell non-cross-listed stocks. Interestingly, when there are small fund flows, there is no evidence of significant reallocation in the post-period. These results suggest that, at least over the timeframe we consider, funds do not fundamentally adjust their country-level allocations as a result of the MMoU. Instead, they shift their within-country holdings toward cross-listed firms, particularly when buying in response to fund inflows or selling in response to fund outflows.

In Column 2, we add time-varying, fund-country, and investee-firm-country fixed effects to control for changes in a fund-country's preference for cross-listings and for the overall preference for cross-listings in each investee-firm country, respectively. In this (much tighter) specification, the coefficient estimates are similar in magnitude to those in Column 1, although the inflow interaction is no longer statistically significant. Taken as a whole, these analyses provide insight into the mechanism by which funds adapt to the MMoU and are consistent with a transactions-cost-based interpretation.

5. Conclusion

Despite the fact that multinational regulatory cooperation is becoming more prevalent, there is little existing empirical evidence examining the effect of changes in regulatory oversight on international investment flows or on regulatory spillovers across countries and investors. We use a country's signing of the MMoU as a shock to cooperation with the US SEC (and a corresponding increase in oversight), coupled with detailed data on portfolio holdings by institutional investors from a wide range of countries, to provide evidence on the effects of cross-border regulatory cooperation on global institutions' portfolio choices.

Our analyses provide what is, to our knowledge, the first evidence of cross-border spillovers from increased regulatory cooperation and information sharing. Specifically, our evidence suggests that additional oversight associated with cooperative regulatory arrangements significantly increases foreign investors' willingness to invest in cross-listed stocks. Our results indicate substantial cross-sectional variation, with the benefits being most pronounced for foreign investors who are at the greatest informational disadvantage absent additional oversight and whose investment strategies are more reliant on information. Although the SEC's focus is primarily on US investors, there are substantial spillover effects for non-US investors and firms. The results also support the bonding hypothesis: stronger US oversight of cross-listed firms leads to an increase in foreign investment in those firms.

While our results are, to some extent, descriptive, they speak to an important gap in the literature and are likely of interest to a range of constituents. First, given that the SEC expends significant effort coordinating with foreign regulators, it is important to understand which investors are the primary beneficiaries. For example, because the SEC is a US regulator, it is interesting to note that the benefits associated with the MMoU appear to be largest for *non-US* investors. Second, the results are likely helpful to signatory countries in weighing the costs and benefits associated with the MMoU. Third, the findings provide insight for stakeholders about the effectiveness of cooperative regulatory arrangements in general and the MMoU in particular.

Of course, our conclusions are subject to caveats. To the extent that countries make other changes that also increase cooperation with the SEC around the time they sign the MMoU [e.g., to facilitate the prosecution of foreign corruption under the Organization for Economic Co-operation and Development's (OECD) Anti-Bribery Convention], the effects we show might not be entirely attributable to the MMoU per se, but rather a function of a broader effort to increase regulatory cooperation. Furthermore, we can only draw inferences from the MMoU as implemented, and much of our evidence on investor, investee, and country characteristics is cross-sectional, limiting our ability to draw strong causal conclusions. That being said, we report a broad set of results that provide insight on regulatory spillovers in global markets.

Appendix A. Variable definitions

Below is a list of variables and their definitions. Subscript i denotes investee firm country, j mutual fund country, t time (year and quarter), and l listing type.

$XLInv_{ijt}$ Aggregate investment in quarter t by funds from country j in firms from country i that cross-list on a US exchange. Cross-listing determination is made from FactSet, with a firm being considered a cross-lister in quarter t if the firm is not a US firm but either has an ADR trading on a US exchange or has an equity issue directly trading on a US exchange.

Inv_{ijt} Aggregate investment in quarter t by funds from country j in firms from country i that either cross-list on a US exchange or do not cross-list in the US.

Inv_{jt} Worldwide mutual fund investments from country j in the stock of firms from the sample countries that either cross-list on a US exchange or do not cross-list in the US.

$XLInvCount_{ijt}$ Aggregate number of investee firms from country i that cross-list on a US exchange invested in by funds from country j during quarter t. Cross-listing determination is made from FactSet, with a firm being considered a cross-lister in quarter t if the firm is not a US firm but either has an ADR trading on a US exchange or has an equity issue directly trading on a US exchange.

$InvCount_{ijt}$ Aggregate number of investee firms from country i that either cross-list on a US exchange or do not cross-list in the US that funds from country j invested in during quarter t.

$PostMMoU_{it}$ Indicator variable equaling one if country i signed the MMoU before quarter t, zero if country i had not applied to sign the MMoU before quarter t, and missing if quarter t is between country i's application to sign the MMoU and the date of signing. If the application date is missing, we assume the mean number of quarters between application and signing (eight quarters). The MMoU signature dates are obtained from IOSCO's website, whereas the application dates were provided by IOSCO directly.

$XLMarketCap_{it}$ Market cap of firms from country i in quarter t that cross-list on a US exchange. Cross-listing determination is made from FactSet, with a firm being considered a cross-lister in quarter t if the firm is not a US firm but either has an ADR trading on a US exchange or has an equity issue directly trading on a US exchange.

$MarketCap_{it}$ Market cap of firms from country i that cross-list on a US exchange or do not cross-list in the US.

$XLMarketCapFreeFloat_{it}$ $XLMarketCap_{it}$ adjusted to exclude closely held shares. Adjustment comes from total shares minus closely held shares, multiplied by year closing price for the average cross-listed firm in Datastream from country i during the year of quarter t as in Dahlquist et al. (2003). As noted by Dahlquist et al. (2003), Datastream's closely held shares data are thinly populated, so to mitigate measurement error, we calculate the free-float adjustment on a country rather than on a firm-level basis.

$MarketCapFreeFloat_{it}$ $MarketCap_{it}$ adjusted to exclude closely held shares. Calculated analogously to $XLMarketCapFreeFloat_{it}$ but using all rather than only cross-listed firms.

$XLFirms_{it}$ Number of firms from country i in quarter t that cross-list on a US exchange, with cross-listing determined as in $XLInvestment_{ijt}$.

$Firms_{it}$ Number of firms from country i that cross-list on a US exchange or do not cross-list in the US.

$NIRatioVW_{it}$ Average net income divided by total assets from firms in country i that cross-list on a US exchange, scaled by average net income over total assets from all firms in country i that either cross-list on a US exchange or have no US cross-listing relationship. Variable calculated on an annual basis. Net income and total asset data come from Worldscope and are the most recent reported from the firm in the past four quarters from quarter t.

$NIRatioEW_{it}$ Calculated analogously as $NIRatioVW_{it}$ but with equal weighting rather than weighting by market capitalization.

$DebtRatioVW_{it}$ Average debt to asset ratio from firms in country i that cross-list on a US exchange, scaled by average debt to asset ratio from all firms in country i that either cross-list on a US exchange or have no US cross-listing relationship, with both the numerator and denominator weighted by market capitalization. Variable calculated on an annual basis. Total debt and total asset data come from Worldscope and are the most recent reported from the firm in the past four quarters from quarter t.

$DebtRatioEW_{it}$ Calculated analogously as $DebtRatioVW_{it}$ but with equal weighting rather than weighting by market capitalization.

$LiqRatioVW_{it}$ Average zero return days per firm of all firms cross-listed on a US exchange from country i during quarter t, weighted by market capitalization, divided by the average zero return days of all firms from country i during quarter t that cross-list on a US exchange or do not cross-list in the US, also weighted by market capitalization. Variable calculated on a quarterly basis. Zero return days are determined from Datastream data.

$LiqRatioEW_{it}$ Calculated analogously as $LiqRatioVW_{it}$ but with equal weighting rather than weighting by market capitalization.

OTC_l Indicator equaling one if the US cross-listing type indicated by group l is sponsored OTC. Zero if the cross-listing type is exchange-listed. To identify sponsored ADRs, we select the CUSIPs of all ADRs in FactSet that are linked to a non-US firm and have the exchange code "PK," and link to the sponsorship in Citigroup's Depositary Receipts directory. For instances where the CUSIP is not in the directory, we manually conduct Google searches on the CUSIP and search for "sponsored" in FactSet's security name field.

$XLInv_{ijtl}$ Aggregate investment in quarter t by funds from country j in firms from country i that have a sponsored OTC US cross-listing if $l =$ OTC or that have an exchange-listed US cross-listing if $l =$ exchange.

Inv_{ijtl} Aggregate investment in quarter t by funds from country j in firms from country i that are either not cross-listed or have US cross-listing (sponsored OTC or exchange-listed) type l.

$LocalFund_{ij}$ Indicator equaling one if $i = j$, zero otherwise.

$Contiguous_{ij}$ Indicator equaling one if i and j share a common border, zero if they do not, and missing if $i = j$. Data obtained from the CEPII GeoDist database.

$SameLanguage_{ij}$ Indicator equaling one if i and j share an official language, zero if they do not, and missing if $i = j$. Language data are obtained from the CEPII GeoDist database.

$HighTrust_{ij}$ Indicator equaling one if on the Eurobarometer survey, a high proportion of respondents from country j state they have "a lot" of trust in people from country i, and zero if the proportion is low. We define a high proportion as being higher than the sample mean ($>$ 20%). Missing if $i = j$, or if i or j are not covered by the Eurobarometer survey (non-European countries).

$SECActivity_i$ Indicator equaling one if country i had a bilateral cooperative agreement with the SEC and the SEC pursued an enforcement action against a firm from country i prior to i's signing the MMoU, and zero otherwise. The bilateral SEC-enforcement-cooperation agreements are obtained from the SEC's website, and SEC enforcement actions are obtained using the procedure in Silvers (2016).

$USTradeImportance_i$ Importance of exports from country i to the US. This variable is calculated by percentile ranking export destinations from country i by export value scaled by counterparty GDP, centered so the mean rank equals zero. Calculated based on trade in the year 2000 from the IMF's Direction of Trade Statistics.

$Transient_s$ Binary variable equaling one if the funds in the observation of category s are labeled as having "high" or "very high" turnover per FactSet's classification. FactSet classifies funds by calculating the average transactions in a year and normalizing by portfolio size. "High" and "very high" classifications correspond to an annual turnover of 100% or greater.

$USFund_j$ Indicator equaling one if country j is the United States, zero otherwise.

$NATO_j$ Indicator equaling one if country j was a member of NATO at the start of the sample, zero otherwise.

$USVotingSim_j$ Percentage of j's voting similarity with the United States in the UN General Assembly in the decade from 1991 through 2000, recentered so the sample mean is zero. UN voting data are obtained from the Harvard Dataverse (Voeten, 2013).

XL_l Indicator if firms of listing type l are cross-listed.

$HiInflow_{jt}$ Indicator if country j's aggregate fund flows in quarter t are in the top quartile of fund flows in the sample. Fund flows are calculated by aggregating each fund's quarterly change in holdings arising from increases or decreases in assets under management that are not driven by price changes in existing investments. Flows are aggregated to the fund-country level and then scaled by total equity invested from the fund country in the quarter for ranking.

$HiOutflow_{jt}$ Indicator if country j's aggregate fund flows in quarter t are in the bottom quartile of fund flows in the sample. Fund flows are calculated as discussed for $HiInflow_{jt}$.

References

Aghion, P., Algan, Y., Cahuc, P., Shleifer, A., 2010. Regulation and distrust. Q. J. Econ. 125 (3), 1015–1049.

Albuquerque, R., Bauer, G.H., Schneider, M., 2009. Global private information in international equity markets. J. Financ. Econ. 94 (1), 18–46.

Bae, K.-H., Stulz, R.M., Tan, H., 2008. Do local analysts know more? A cross-country study of the performance of local analysts and foreign analysts. J. Financ. Econ. 88 (3), 581–606.

Baik, B., Kang, J.-K., Kim, J.-M., 2010. Local institutional investors, information asymmetries, and equity returns. J. Financ. Econ. 97 (1), 81–106.

Bekaert, G., Harvey, C.R., Lumsdaine, R.L., 2002. Dating the integration of world equity markets. J. Financ. Econ. 65 (2), 203–247.

Benos, E., Weisbach, M.S., 2004. Private benefits and cross-listings in the United States. Emerg. Markets Rev. 5 (2), 217–240.

Bernile, G., Kumar, A., Sulaeman, J., 2014. Leveling the playing field: financial regulation and disappearing local bias of institutional investors. Singapore Management University and University of Miami. Unpublished working paper

Borochin, P., Yang, J., 2017. The effects of institutional investor objectives on firm valuation and governance. J. Financ. Econ. 126 (1), 171–199.

Bushee, B.J., 1998. The influence of institutional investors on myopic R&D investment behavior. Account. Rev. 73, 305–333.

Bushee, B.J., 2001. Do institutional investors prefer near-term earnings over long-run value? Contemp. Account. Res. 18 (2), 207–246.

Carlin, B.I., Dorobantu, F., Viswanathan, S., 2009. Public trust, the law, and financial investment. J. Financ. Econ. 92 (3), 321–341.

Christensen, H.B., Hail, L., Leuz, C., 2016. Capital-market effects of securities regulation: Prior conditions, implementation, and enforcement. Rev. Financ. Stud. 29 (11), 2885–2924.

Christensen, H.B., Maffett, M.G., Rauter, T., 2019. Policeman for the world: Us enforcement of foreign corruption regulation and corporate investment policies. University of Chicago Booth School of Business. Unpublished working paper

Coffee, J.C., 1998. Future as history: the prospects for global convergence in corporate governance and its implications, 93. Northwestern University Law Review, p. 641.

Dahlquist, M., Pinkowitz, L., Stulz, R.M., Williamson, R., 2003. Corporate governance and the home bias. J. Financ. Quant. Anal. 38 (1), 87–110.

Doidge, C., 2004. US cross-listings and the private benefits of control: evidence from dual-class firms. J. Financ. Econ. 72 (3), 519–553.

Doidge, C., Karolyi, G.A., Stulz, R.M., 2004. Why are foreign firms listed in the US worth more? J. Financ. Econ. 71 (2), 205–238.

Eleswarapu, V.R., Venkataraman, K., 2006. The impact of legal and political institutions on equity trading costs: a cross-country analysis. Rev. Financ. Stud. 19 (3), 1081–1111.

Fang, V.W., Maffett, M., Zhang, B., 2015. Foreign institutional ownership and the global convergence of financial reporting practices. J. Account. Res. 53 (3), 593–631.

Ferreira, M.A., Matos, P., 2008. The colors of investors' money: the role of institutional investors around the world. J. Financ. Econ. 88 (3), 499–533.

Ferreira, M.A., Matos, P., Pereira, J.P., Pires, P., 2017. Do locals know better? A comparison of the performance of local and foreign institutional investors. J. Bank. Finance 82, 151–164.

Guiso, L., Sapienza, P., Zingales, L., 2009. Cultural biases in economic exchange? Quart. J. Econ. 124 (3), 1095–1131.

Hail, L., Leuz, C., 2006. International differences in the cost of equity capital: do legal institutions and securities regulation matter? J. Account. Res. 44 (3), 485–531.

Jagannathan, M., Jiao, W., Karolyi, G. A., 2017. Is there a home field advantage in global markets?Unpublished working paper. Cornell University.

Karolyi, G.A., 1998. Why do companies list shares abroad?: A survey of the evidence and its managerial implications. Financ. Mark. Inst. Instrum. 7 (1), 1–60.

Kho, B.-C., Stulz, R.M., Warnock, F.E., 2009. Financial globalization, governance, and the evolution of the home bias. J. Account. Res. 47 (2), 597–635.

Leuz, C., Lins, K.V., Warnock, F.E., 2009. Do foreigners invest less in poorly governed firms? Rev. Financ. Stud. 22 (8), 3245–3285.

Leuz, C., Wysocki, P.D., 2016. The economics of disclosure and financial reporting regulation: Evidence and suggestions for future research. J. Account. Res. 54 (2), 525–622.

Licht, A.N., 1998. International diversity in securities regulation: Roadblocks on the way to convergence. Cardozo Law Rev. 20, 227–286.

Miller, D.P., 1999. The market reaction to international cross-listings: evidence from depositary receipts. J. Financ. Econ. 51 (1), 103–123.

Mulherin, J.H., 2007. Measuring the costs and benefits of regulation: Conceptual issues in securities markets. J. Corporate Finance 13 (2–3), 421–437.

Pinotti, P., 2008. Trust, Honesty, and Regulations. Unpublished working paper. Bank of Italy and Universitat Pompeu Fabra.

Siegel, J., 2005. Can foreign firms bond themselves effectively by renting us securities laws? J. Financ. Econ. 75 (2), 319–359.

Silvers, R., 2016. The valuation impact of sec enforcement actions on nontarget foreign firms. J. Account. Res. 54 (1), 187–234.

Silvers, R., 2018. Cross-border cooperation between securities regulators. University of Utah doi:10.1016/j.jacceco.2020.101301.

Stulz, R.M., 1999. Globalization, corporate finance, and the cost of capital. J. Appl. Corporate Finance 12 (3), 8–25.

Voeten, E., 2013. Data and analyses of voting in the United Nations General Assembly. In: Reinalda, B. (Ed.), Routledge Handbook of International Organization. Routledge, pp. 80–92.